

POSTS AND TELECOMMUNICATIONS INSTITUTE OF TECHNOLOGY
FACULTY OF INFORMATION TECHNOLOGY 1



MAJOR ASSIGNMENT
ANALYSIS AND DESIGN OF INFORMATION SYSTEMS
SUBJECT 30

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A garage management system enables management staff, sales staff, warehouse staff, and customers to use:

- Management staff: view the statistics categories: services, parts, customers and suppliers. Service information management, spare parts.
- Warehouse staff: import spare parts from suppliers, manage supplier information
- Sale staff: receive customers, receive service and spare parts requests from customers, assign technical staff according to customer service requirements, receive payment from customers.
- Customer: Search service information, book an appointment online

Considering two modules and answering questions

- ***Customer searches for service information:*** selects the menu to find service / spare parts information → enters the service / part name to search → the system displays a list of services / parts whose name contains the entered keyword → clicks on a service/part to view details → the system displays detailed information about the service/part.
- ***Management staff views customer statistics:*** selects the menu to view statistical reports → selects customer statistics by revenue → selects the start and end date of statistics → views customer statistics → clicks on a customer → view details of the invoices of the customer → clicks on an invoice → view the detail selected invoice.

Question 1 (2 points)

- a. Build the use case diagram for the two modules
- b. Present scenarios for the two modules

Question 2 (2 points)

- a. Extract the entity classes (class name, attributes) related to the two modules
- b. Build the entity class diagram for the extracted classes.

Question 3 (2 points)

- a. Build the communication diagram for the standard scenario of the two modules.
- b. Build the detail design class diagram for the two modules.

Question 4 (2 points)

- a. Based on the entity classes of Question 2, design the database related to the two modules
- b. Generate Java code (class template, declaration of attributes and methods, explain how methods work) for classes in the diagram of Question 3.b.

Question 5 (2 points)

- a. Build the package diagram for classes in the diagram of Question 3.b.
- b. Build the deployment diagram for MVC model based on the J2EE technology.

I. Requirement

1. Build the glossary list

Order	Name	Meaning	Type
1	Customer	Person who use the system to search service and spare part information by enter keyword, book an appointment online, and make payments for received services and spareparts. Customer interacts directly with the system through a web interface to conveniently manage their repair and maintenance needs.	Human
2	Management Staff	Employees who responsible for overseeing the overall operation of the garage. They use the system to view detailed statistical reports related to services, spare parts, customers and suppliers. They also manage and update service and spare part information to ensure accuracy and completeness.	Human
3	Warehouse Staff	Employees in charge of maintaining and updating inventory records. They use the system to import new spare parts from suppliers, track stock levels, and manage supplier details such as contact information and transaction history.	Human
4	Sales Staff	Employees who interact directly with customers. They use the system to receive service and spare parts requests, assign technical staff based on customer requirements, and handle billing and payment processes to ensure smooth transactions.	Human
5	Search Service Information	The process through which a customer enters a keyword to find information about available services offered by the garage. This includes viewing a list of services matching the search keyword and selecting specific services for more details.	Action of Actor

6	Search Spare Parts Information	The process through which a customer enters a keyword to find information about available spare parts offered by the garage. This includes viewing a list of spare parts matching the search keyword and selecting specific spare parts for more details.	Action of Actor.
7	Book an appointment online	The process when a customer schedules a service visit via the online systems. The system records the customer's preferred date, time, and type of service, spare parts, helping the garage manage service schedules efficiently.	Action of Actor
8	Enter Keyword	The customer's action of typing a service or spare part name into a search field to retrieving relevant information. The entered keyword is processed by the system to generate matching results.	Action of Actor
9	View Service Details	The customer's action of selecting a specific service from the search result list to see complete details, such as description, price, estimated duration, and any special conditions.	Action of Actor
10	View Spare Part Details	The customer's action of selecting a particular spare part from the search result list to view information such as part name, model number, compatibility, stock availability, and price.	Action of Actor
11	View the statistics categories	The action of the management staff choosing which statistical report to view among categories like services, spare parts, customers, and suppliers. These statistics help management assess business performance and make strategic decisions.	Action of Actor

12	Manage Service Information	The management staff's activity of adding, updating, or removing information related to the garage's services. This may include editing service names, descriptions, pricing to maintain data accuracy.	Action of Actor
13	Manage Spare Part Information	The management staff's task of maintaining accurate spare part information in the system, including stock status, supplier details, and pricing. It ensures that the system always reflects current inventory levels.	Action of Actor
14	View Customer Statistics	The management staff's action of selecting a statistical report type (e.g., revenue by customer) to analyze customer behavior, total spending, and service usage patterns over a given time period.	Action of Actor
15	Select Revenue Statistics	The management staff's action of choosing to view statistics focused on customer revenue generation within a defined date range, allowing analysis of business profitability and customer contribution.	Action of Actor
16	View Invoice Details	The action of the management staff selecting a particular invoice from a customer's transaction history to view detailed billing information, including services rendered, spare parts used, total cost, and payment status.	Action of Actor
17	View Customer Details	The action of management staff selecting a customer from the statistics list to access their detailed transaction history, contact information, and associated invoices.	Action of Actor
18	Import Spare Parts	The process by which warehouse staff record new spare part entries	Action of Actor

		from suppliers into the system. This includes inputting part details, quantities, supplier names, and import dates to maintain accurate stock records.	
19	Manage Supplier Information	The warehouse staff's action of updating supplier data in the system, such as company name, contact details, supplied parts list, and past transactions. This helps ensure effective supplier relationship management.	Action of Actor
20	Receive Customer	The action of sales staff welcoming customers to the garage, gathering necessary information about their vehicles and desired services, and entering these details into the system for record-keeping.	Action of Actor
21	Receive service and spare parts	The activity of sales staff recording customer requests for specific services or spare parts. These requests are later processed by assigning technical staff or checking stock availability.	Action of Actor
22	Assign Technical Staff	The sales staff's task of designating appropriate technical personnel to handle a customer's service based on skill, availability, and service type.	Action of Actor
23	Receive Payment	The action of sales staff processing the customer's payment for completed services or purchased spare parts. This includes updating the system with payment details and generating a receipt or invoice.	Action of Actor
24	Service	A specific maintenance or repair task provided by the garage, such as oil changes, engine tuning, or tire replacement. Each service has	Object

		attributes such as name, description, cost.	
25	Spare Part	A mechanical or electrical component used for vehicle maintenance or repair, typically supplied by vendors. Each spare part includes attributes like part ID, name, unit price, and stock quantity.	Object
26	Invoice	A formal document generated by the system that records details of customer transactions, including services provided, spare parts sold, and payment amounts. It serves as proof of payment and supports financial tracking.	Object
27	Supplier	A company or individual that provides spare parts or equipment to the garage. Supplier records include company details, contact information, and lists of supplied parts.	Object
28	Statistics	A summarized report produced by the system that presents data on services, spare parts, customers, or suppliers to support managerial analysis and decision-making.	Object
29	Appointment	A scheduled service booking made by a customer through the online system. It includes details such as the chosen service, appointment date, and customer contact information.	Object
30	Customer Statistics	A specialized statistical report that summarizes the total revenue generated by each customer within a given period, helping management evaluate customer value and business performance.	Object

31	Display Service/Part List	The system's function of displaying all services or spare parts whose names match the entered keyword, allowing the customer to browse through options before selecting one.	Action of System
32	Display Service/Part Details	The system's response action of showing comprehensive details about a selected service or spare part, such as description, price, and availability.	Action of System
33	View Statistics List	The system's function of presenting summarized statistical data on customer revenue or other metrics based on the selected criteria and date range.	Action of System
34	View Invoice List	The system's process of displaying all invoices associated with a selected customer to help management review their transaction history.	Action of System
35	Display Invoice Details	The system's function of showing complete details of a chosen invoice, including service names, quantities, amounts, and payment dates.	Action of System
36	System	The Garage Management System (GMS) is an integrated software solution that supports garage operations such as customer service management, spare parts inventory, supplier coordination, and business analytics. It connects customers and staff through a centralized database and user-friendly interface.	System

2. Project information

a. Objective

- Building a garage management system that supports many types of users: including management staff, warehouse staff, sales staff and customers.
- Allow customers to easily search for services/parts and book appointments online.

- Enable management staff to monitor and generate reports on business performance (services, parts, customers, suppliers).
- Support sales staff and warehouse staff in handling their daily tasks (importing parts, managing suppliers, processing payments).
- Improve transparency, save time, and enhance customer experience.

b. Scope

Main Users:

- **Management staff:**
 - View the statistics categories: services, parts, customers and suppliers.
 - Manage service information, spare parts.
- **Sales staff:**
 - Receive customers, service/part requests from customers.
 - Assign technicians
 - Handle customer payments.
- **Warehouse staff:**
 - Import spare parts from suppliers
 - Manage supplier information.
- **Customers:**
 - Search for service/part information
 - Book appointments online.

Key Modules (two selected according to the requirement):

1. **Customer searches for service/spare parts information**
2. **Management staff views customer statistics**

c. How do the modules work

Module 1: Customer searches for service/spare parts information

1. The customer selects the menu to search for services / spare parts information.
2. The customer enters the name of the service/part.
3. The system displays a list of services/parts matching the entered keyword.
4. The customer clicks on a service/part to view details.
5. The system displays detailed information about the service/part.

Module 2: Management staff views customer statistics

1. Management staff selects the menu to view statistical reports.
2. Management staff chooses customer statistics by revenue.
3. Management staff selects the start and end date of the report.
4. The system displays customer revenue statistics.
5. Management staff clicks on a specific customer → system shows the customer's invoices.

6. Management system clicks on a specific invoice → system displays detailed invoice information.

d. Information about objects

- Member: id, username, password, name, dateOfBirth, address, email, phone, role, notes
- Staff: position
- ManagementStaff: position
- WarehouseStaff: position
- SaleStaff: position
- TechnicalStaff: position
- Customer: idCustomer
- Address: id, homeNumber, roadName, district, province, nation
- Vehicle: id, customer, licensePlate, desc
- AppointmentOnline: id, customer, date, time, status, desc
- ReceivingSlip: id, customer, vehicle, time, totalAmount, status, saleStaff
- PaymentInvoice: id, time, receivingSlip, totalAmount
- ServiceSlip: id, service, quantity, totalAmount, technicalStaff
- SparePartSlip: id, sparePart, quantity, totalAmount
- Service: id, name, price, desc
- SparePart: id, name, price, quantity, desc
- ImportInvoice: id, time, totalAmount, status, desc, warehouseStaff, supplier
- ImportSparePart: id, importInvoice, sparePart, quantity, totalAmount, desc
- Supplier: id, name, phone, email, address, desc
- CustomerStat: revenue, number, startDate, endDate
- ServiceStat: totalAmount, revenue, quantity, startDate, endDate
- SparePartStat: totalAmount, revenue, quantity, startDate, endDate
- SupplierStat: revenue, startDate, endDate

e. Relationships among objects

- **Customer – AppointmentOnline**
A customer can create multiple online appointments; each appointment belongs to one customer. (1–N)
- **Customer – Vehicle**
Each customer can own multiple vehicles, while every vehicle belongs to one customer. (1–N)
- **Customer – ReceivingSlip**
A customer can have many receiving slip; each receiving slip is linked to one customer. (1–N)
- **Vehicle – ReceivingSlip**
A vehicle may appear in several receiving slip (for different services); each receiving slip corresponds to a specific vehicle. (1–N)

- **ReceivingSlip – ServiceSlip**
Each receiving slip can include multiple service slips; each service slip is connected to a single receiving slip. (1–N)
- **ReceivingSlip – SparePartSlip**
Each receiving slip may contain several spare part slips; every spare part slip belongs to one receiving slip. (1–N)
- **ServiceSlip – Service**
A service slip refers to one specific service; a service can appear in many slips. (N–1)
- **SparePartSlip – SparePart**
Each spare part slip refers to one spare part; one spare part can be included in many slips. (N–1)
- **SparePart – Supplier**
Each spare part is supplied by one supplier; a supplier can provide multiple spare parts. (1–N)
- **ImportInvoice – ImportSparePart**
An import invoice can contain several imported spare parts; each imported spare part record belongs to one import invoice. (1–N)
- **ImportSparePart – SparePart**
Each imported spare part record corresponds to a single spare part; one spare part can appear in multiple import records. (N–1)
- **ImportInvoice – WarehouseStaff**
Each import invoice is handled by one warehouse staff member; a warehouse staff member can manage multiple import invoices. (1–N)
- **ReceivingSlip– SaleStaff**
Each receiving slip is processed by one sales staff member; one sales staff member can manage many receiving slips. (1–N)
- **ServiceSlip – TechnicalStaff**
Each service slip is carried out by one technical staff member; one technical staff member can perform multiple service slips. (1–N)
- **ManagementStaff – Report (via Statistics Classes)**
Management staff can generate multiple statistical reports (such as customer, service, spare part, and supplier statistics). Each report summarizes aggregated data related to the system's operations.
- **Staff – Member (Inheritance)**
Staff is a subclass of Member. Different types of staff (ManagementStaff, WarehouseStaff, SaleStaff, TechnicalStaff) inherit attributes and behaviors from Staff. (Inheritance)
- **ReceivingSlip - PaymentInvoice**
Each receiving slip connect one payment invoice. (1–1)
- **Statistical Classes (CustomerStat, ServiceStat, SparePartStat, SupplierStat)**
Each statistical class represents aggregated data (such as total revenue,

quantity, and performance) for its related entity. These classes maintain **1–1** or **1–N** conceptual relationships depending on the reporting time frame.

3. Draw Use case diagrams

a. General use case diagram

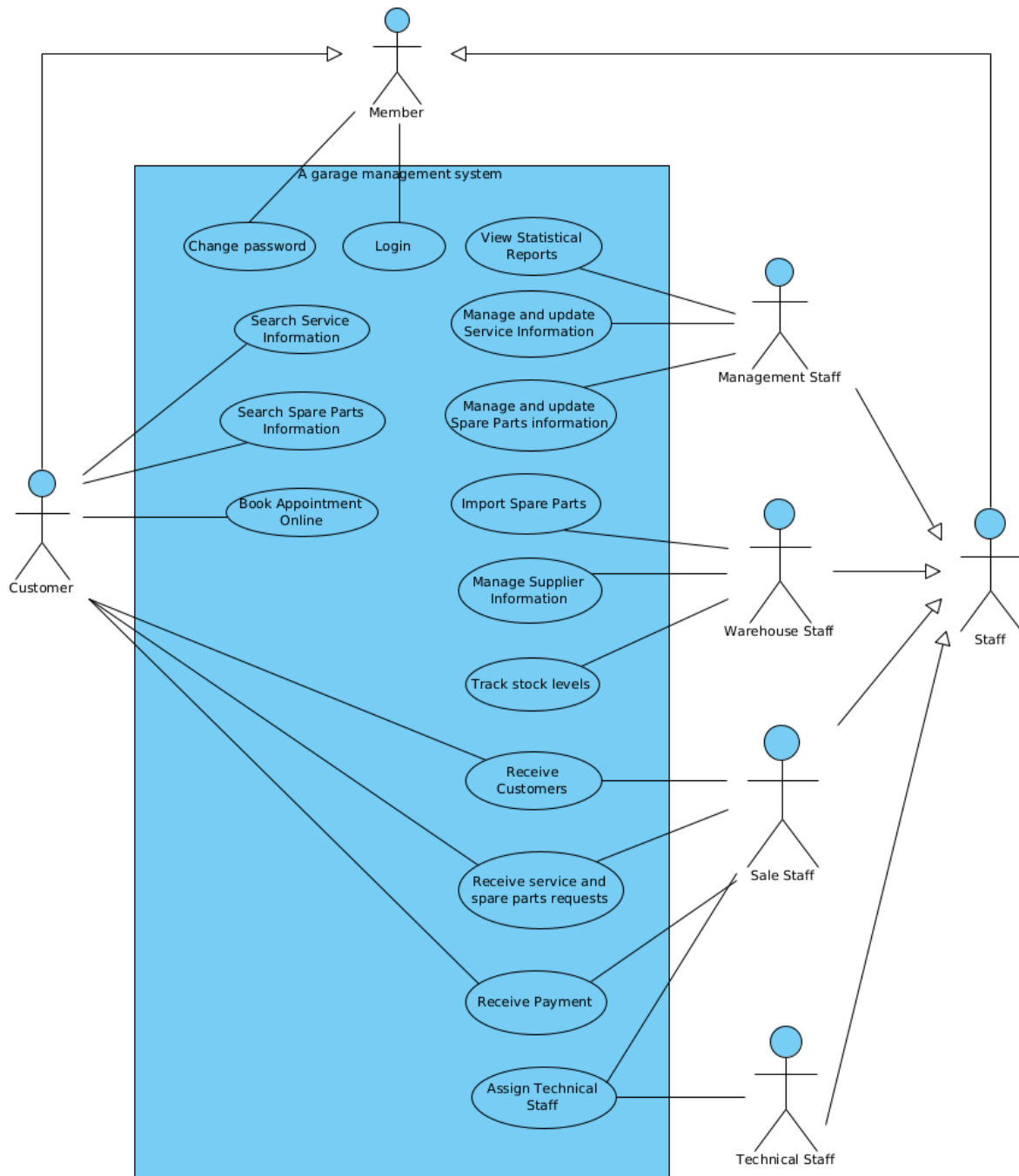


Figure 1.1. General use case diagram

b. Detail use case diagram

Module 1: Customer searches for service/spare parts information

1. The customer selects the menu to search for services / spare parts information.
2. The customer enters the name of the service/part.
3. The system displays a list of services/parts matching the entered keyword.
4. The customer clicks on a service/part to view details.
5. The system displays detailed information about the service/part.

Usecase diagram:

Customer searches for service information

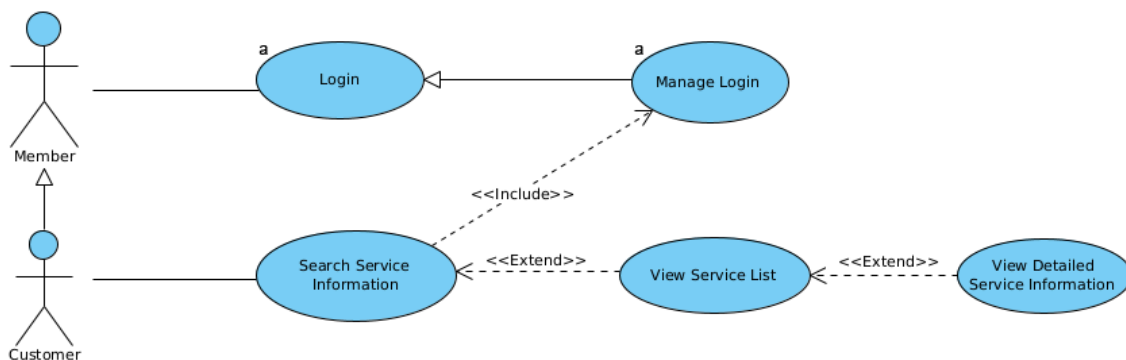


Figure 1.2. Detailed usecase diagram for service search module

Customer searches for spare parts information

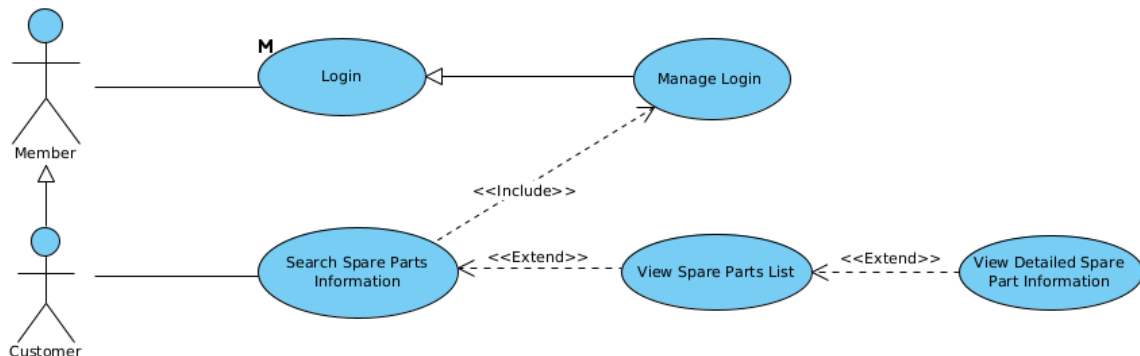


Figure 1.3. Detailed usecase diagram for spare parts search module

Description: This use case describes how a customer can search for service or spare part information in the system by entering a keyword, viewing the list of results, and checking detailed information of a selected item.

Main Flow:

1. The customer selects the option Search Service/Spare Parts Information.
2. The system prompts the customer to Enter Service/Spare Parts Name.

3. The customer enters the name (or part of the name) of the desired service/spare part.
4. The system retrieves and displays a Service/Spare Parts list that matches the entered keyword.
5. The customer selects a specific item from the list.
6. The system displays detailed information about the selected service/spare part.

Module 2: Management staff views customer statistics

1. Management staff selects the menu to view statistical reports.
2. Management staff chooses customer statistics by revenue.
3. Management staff selects the start and end date of the report.
4. The system displays customer revenue statistics.
5. Management staff clicks on a specific customer → system shows the customer's invoices.
6. Management system clicks on a specific invoice → system displays detailed invoice information.

Usecase diagram:

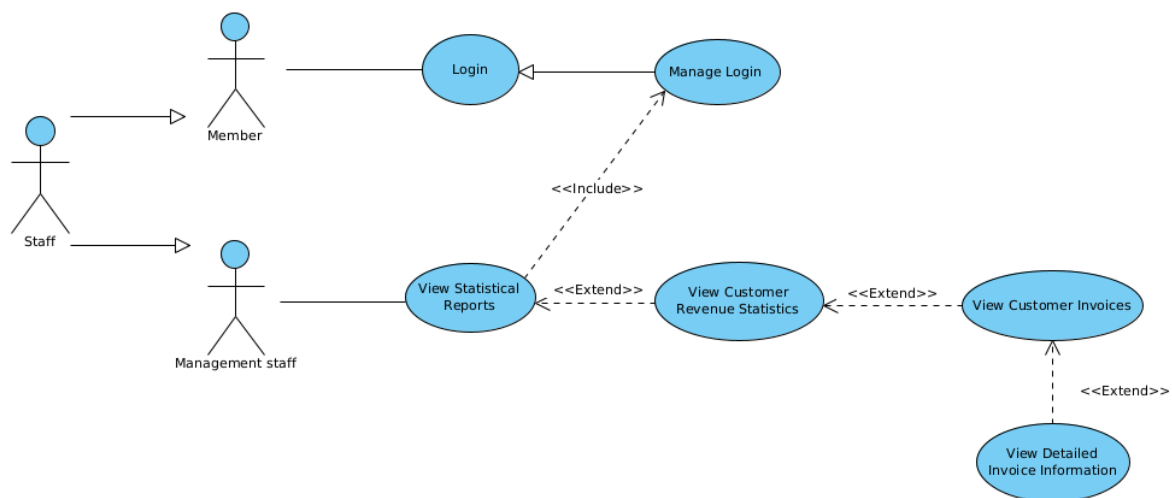


Figure 1.4. Detailed usecase diagram for viewing customer statistics

Description: This use case describes how the management staff can view statistical reports of the system, including customer revenue statistics and detailed invoice information.

Main Flow:

1. The management staff selects the option View Statistical Reports.

2. The system provides access to different types of reports.
3. The management staff chooses View Customer Revenue Statistics (extends use case).
4. The system requests the staff to Select Date Range.
5. The management staff enters a specific date range.
6. The system processes the request and generates revenue statistics for the selected period.
7. The staff may also view related invoices by selecting View Customer Invoices.
8. The system displays a list of invoices for the selected customers and date range.
9. The management staff selects a specific invoice.
10. The system displays Detailed Invoice Information.

II. Analysis

1. Write scenarios for 2 modules.

Module 1: Customer searches for service/spare parts information

1. The customer selects the menu to search for services / spare parts information.
2. The customer enters the name of the service/part.
3. The system displays a list of services/parts matching the entered keyword.
4. The customer clicks on a service/part to view details.
5. The system displays detailed information about the service/part.

Use case	Search for service information																								
Actor	Customer																								
Precondition	The customer has successfully logged in (if required). The system is available, responsive, and connected to the service database.																								
Postcondition	The customer successfully views detailed information about the selected service, or receives a clear message explaining why the search could not be completed.																								
Main scenario	1. From the main interface, the customer selects the menu option to search for services. 2. The system displays the search interface with an input field for keywords. 3. The customer enters the name (or partial name) of the desired service. 4. The system displays a list of matching services. Find Service Information Service Catalog change Search Search Spare Parts <table><tr><th>ID</th><th>NAME</th><th>PRICE (VND)</th><th>ACTION</th></tr><tr><td>S1</td><td>Oil Change</td><td>200,000</td><td>View Details</td></tr><tr><td>S8</td><td>Transmission Fluid Change</td><td>450,000</td><td>View Details</td></tr><tr><td>S20</td><td>Power Steering Fluid Change</td><td>240,000</td><td>View Details</td></tr><tr><td>S30</td><td>Differential Fluid Change</td><td>380,000</td><td>View Details</td></tr><tr><td>S53</td><td>Clutch Fluid Change</td><td>290,000</td><td>View Details</td></tr></table> Back to Home 5. The customer selects one service from the list.	ID	NAME	PRICE (VND)	ACTION	S1	Oil Change	200,000	View Details	S8	Transmission Fluid Change	450,000	View Details	S20	Power Steering Fluid Change	240,000	View Details	S30	Differential Fluid Change	380,000	View Details	S53	Clutch Fluid Change	290,000	View Details
ID	NAME	PRICE (VND)	ACTION																						
S1	Oil Change	200,000	View Details																						
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S20	Power Steering Fluid Change	240,000	View Details																						
S30	Differential Fluid Change	380,000	View Details																						
S53	Clutch Fluid Change	290,000	View Details																						

	6. The system displays detailed information about the selected service, including description, price, and other relevant details. Find Service Information Service Detail ID: S1 Name: Oil Change Price: 200,000 VND Description: Engine oil replacement Back to List Home
Exceptions	3. The customer leaves the search field empty and clicks "Search". The system displays a message: <i>"Please enter a keyword before searching."</i> 3. The customer enters invalid characters (e.g., special symbols or emojis). The system notifies the customer: <i>"Invalid characters detected. Please enter letters or numbers only."</i> 4. No service matches the entered keyword → the system notifies the customer: <i>"No results found. Please try again with a different keyword."</i> 4. The database connection fails during the search. The system displays: <i>"Unable to connect to the database. Please try again later."</i> 4. The system takes too long to retrieve the results (timeout). The system notifies: <i>"Search timed out. Please check your internet connection and try again."</i> 6. The customer clicks on a service, but the data record has been deleted or is corrupted. The system displays: <i>"This service is no longer available. Please select another item."</i> 6. The service details page fails to load completely due to a system error. The system shows: <i>"Error loading service details. Please refresh or try again later."</i>

Use case	Search for spare parts information																														
Actor	Customer																														
Precondition	The customer has successfully logged in (if required). The system is available, responsive, and connected to the spare parts database.																														
Postcondition	The customer successfully views detailed information about the selected spare part, or receives a clear and appropriate message explaining why the search could not be completed.																														
Main scenario	1. From the main interface, the customer selects the menu option to search for spare parts. 2. The system displays the spare part search interface with an input field for keywords. 3. The customer enters the name (or partial name) of the desired spare part. 4. The system searches the database and displays a list of spare parts whose names match the entered keyword. Find Spare Part Information Spare Part Catalog oil Search Search Services <table><tr><th>ID</th><th>NAME</th><th>PRICE (VND)</th><th>STOCK</th><th>ACTION</th></tr><tr><td>SP1</td><td>Engine Oil</td><td>100,000</td><td>100 pcs</td><td>View Details</td></tr><tr><td>SP7</td><td>Oil Filter</td><td>120,000</td><td>90 pcs</td><td>View Details</td></tr><tr><td>SP33</td><td>Ignition Coil</td><td>580,000</td><td>42 pcs</td><td>View Details</td></tr><tr><td>SP46</td><td>Transmission Oil</td><td>180,000</td><td>80 pcs</td><td>View Details</td></tr><tr><td>SP47</td><td>Differential Oil</td><td>220,000</td><td>70 pcs</td><td>View Details</td></tr></table> Back to Home 5. The customer reviews the list and selects one spare part from the results. 6. The system displays detailed information about the selected spare part, including part name, stock quantity, price.	ID	NAME	PRICE (VND)	STOCK	ACTION	SP1	Engine Oil	100,000	100 pcs	View Details	SP7	Oil Filter	120,000	90 pcs	View Details	SP33	Ignition Coil	580,000	42 pcs	View Details	SP46	Transmission Oil	180,000	80 pcs	View Details	SP47	Differential Oil	220,000	70 pcs	View Details
ID	NAME	PRICE (VND)	STOCK	ACTION																											
SP1	Engine Oil	100,000	100 pcs	View Details																											
SP7	Oil Filter	120,000	90 pcs	View Details																											
SP33	Ignition Coil	580,000	42 pcs	View Details																											
SP46	Transmission Oil	180,000	80 pcs	View Details																											
SP47	Differential Oil	220,000	70 pcs	View Details																											

	Find Spare Part Information Spare Part Detail ID: SP1 Name: Engine Oil Price: 100,000 VND Stock: 100 pcs Description: Synthetic 5W-30 oil Back to List Home
Exceptions	3. The customer leaves the search field empty and clicks “Search”. The system displays a message: <i>“Please enter a keyword before searching.”</i> 3. The customer enters invalid characters (e.g., special symbols or emojis). The system notifies the customer: <i>“Invalid characters detected. Please enter letters or numbers only.”</i> 4. No spare part matches the entered keyword → the system notifies the customer: <i>“No results found. Please try again with a different keyword.”</i> 4. The database connection fails during the search. The system displays: <i>“Unable to connect to the database. Please try again later.”</i> 4. The system takes too long to retrieve the results (timeout). The system notifies: <i>“Search timed out. Please check your internet connection and try again.”</i> 6. The customer clicks on a spare part, but the data record has been deleted or is corrupted. The system displays: <i>“This spare part is no longer available. Please select another item.”</i> 6. The spare part details page fails to load completely due to a system error. The system shows: <i>“Error loading spare part details. Please refresh or try again later.”</i>

Module 2: Management staff views customer statistics

1. Management staff selects the menu to view statistical reports.
2. Management staff chooses customer statistics by revenue.
3. Management staff selects the start and end date of the report.
4. The system displays customer revenue statistics.

5. Management staff clicks on a specific customer → system shows the customer's invoices.
6. Management system clicks on a specific invoice → system displays detailed invoice information.

Use case	View customer statistics																																																																						
Actor	Management staff																																																																						
Precondition	The management staff has successfully logged in (if required). The system is available, responsive, and connected to the database.																																																																						
Postcondition	The management staff successfully views detailed customer revenue statistics, or receives a clear error message if the process fails.																																																																						
Main scenario	1. From the main interface, the management staff selects the menu to view statistics. 2. The system displays types of statistical reports. 3. The management staff chooses “Customer Statistics by Revenue”. 4. The system displays the start and end date selection form. 5. The management staff selects the start and end date of the report. 6. The system displays customer revenue statistics for the selected period. <table><tr><th>Customer ID</th><th>Name</th><th>Invoices</th><th>Revenue (VND)</th><th>Action</th></tr><tr><td>M2</td><td>Nguyen Huu Phuc</td><td>1</td><td>950,000</td><td>View Invoices</td></tr><tr><td>M8</td><td>Nguyen Van Hung</td><td>1</td><td>820,000</td><td>View Invoices</td></tr><tr><td>M23</td><td>Trinh Hoai Nam</td><td>1</td><td>720,000</td><td>View Invoices</td></tr><tr><td>M17</td><td>Dang Thi Lan</td><td>1</td><td>660,000</td><td>View Invoices</td></tr><tr><td>M20</td><td>Ngo Thanh Thuy</td><td>1</td><td>300,000</td><td>View Invoices</td></tr><tr><td>M14</td><td>Le My Hoa</td><td>1</td><td>150,000</td><td>View Invoices</td></tr><tr><td>M10</td><td>Le Van Hoang</td><td>1</td><td>100,000</td><td>View Invoices</td></tr></table> Total Revenue: 3,700,000 VND 7. The management staff clicks on a specific customer. 8. The system displays the customer’s invoices. <table><tr><th>ID</th><th>Date</th><th>Total Amount (VND)</th><th>Type</th><th>Action</th></tr><tr><td>PI1</td><td>01/10/2025</td><td>1,180,000</td><td>Card</td><td>View</td></tr><tr><td>PI7</td><td>04/10/2025</td><td>1,530,000</td><td>Bank Transfer</td><td>View</td></tr><tr><td>PI20</td><td>09/10/2025</td><td>150,000</td><td>Cash</td><td>View</td></tr><tr><td>PI31</td><td>01/11/2025</td><td>950,000</td><td>Cash</td><td>View</td></tr><tr><td>PI41</td><td>17/11/2025</td><td>410,000</td><td>Cash</td><td>View</td></tr></table> 9. The management staff clicks on a specific invoice.	Customer ID	Name	Invoices	Revenue (VND)	Action	M2	Nguyen Huu Phuc	1	950,000	View Invoices	M8	Nguyen Van Hung	1	820,000	View Invoices	M23	Trinh Hoai Nam	1	720,000	View Invoices	M17	Dang Thi Lan	1	660,000	View Invoices	M20	Ngo Thanh Thuy	1	300,000	View Invoices	M14	Le My Hoa	1	150,000	View Invoices	M10	Le Van Hoang	1	100,000	View Invoices	ID	Date	Total Amount (VND)	Type	Action	PI1	01/10/2025	1,180,000	Card	View	PI7	04/10/2025	1,530,000	Bank Transfer	View	PI20	09/10/2025	150,000	Cash	View	PI31	01/11/2025	950,000	Cash	View	PI41	17/11/2025	410,000	Cash	View
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PI41	17/11/2025	410,000	Cash	View																																																																			

	10. The system displays detailed invoice information.
Exceptions	2. The system fails to load the list of report types. The system displays: <i>"Unable to load report types. Please refresh or try again later."</i> 3. The selected report type is not available or has been removed. The system displays: <i>"The selected report type is currently unavailable. Please choose another option."</i> 4. The date selection form fails to load. The system displays: <i>"Error loading date selection. Please refresh or try again later."</i> 5. The management staff leaves one or both date fields empty. The system displays: <i>"Please select both start and end dates before generating the report."</i> 5. The end date is earlier than the start date. The system displays: <i>"End date must be later than start date."</i> 6. No data is available for the selected period. The system displays: <i>"No data found for the selected period."</i> 6. Database connection fails during data retrieval. The system displays: <i>"Unable to connect to the database. Please try again later."</i> 6. Report generation times out due to system overload. The system displays: <i>"Report generation timed out. Please try again later."</i> 7. The management staff clicks on a customer whose record was deleted. The system displays: <i>"Customer record not found. Please refresh the report."</i> 9. The management staff clicks on an invoice that was deleted or corrupte. The system displays: <i>"Invoice data unavailable. Please select another record."</i> 10. The invoice details page fails to load completely due to a system error. The system displays: <i>"Error loading invoice details. Please refresh or try again later."</i>

2. Construct the entity class diagram.

Step 1: Describe the system in one paragraph

The garage management system is a website that supports the management of a garage and allows different types of users to perform their tasks. In there, management staff is allowed to view the service statistics, spare parts statistics, customers statistics and suppliers statistics; manage information service and spare parts. Warehouse staff are allowed to import spare parts from suppliers; manage supplier information. Sales staff are allowed to receive customers; receive service and spare parts that request from customers throw receiving slip; assign technical staff according to customer service requirements; and receive payment from customers. Customers are allowed to search service and spare parts information; book an appointment online.

Step 2: Extract nouns from the paragraph

- Humans: Staff, ManagementStaff, WarehouseStaff, SaleStaff, TechnicalStaff, Customer, Supplier
- Objects: Vehicle, SparePart, Service, AppointmentOnline, Receiving Slip, ImportInvoice, PaymentInvoice, ServiceSlip, SparePartSlip
- Statistics: ServiceStat, SparePartStat, CustomersStat, SupplierStat

Step 3: Evaluate and choose nouns to do entity class or attribute

- Member (abstract): username, password, name, dateofbirth, address, email, phone, role, notes
- Staff (abstract): position
- ManagementStaff
- WarehouseStaff
- SaleStaff
- TechnicalStaff
- Customer: idCustomer
- Vehicle: licencePlate, desc
- Address: homenummer, roadname, district, province, nation
- Supplier: name, address, phone, email, desc
- SparePart: name, price, quantity, desc
- Service: name, price, desc
- AppointmentOnline: date, time, status, desc
- ImportInvoice: time, totalAmount, status, desc
- ImportSparePart: quantity, totalAmount, desc
- ReceivingSlip: time, totalAmount
- PaymentInvoice: time, totalAmount, type
- ServiceSlip: quantity, totalAmount
- SparePartSlip: quantity, totalAmount
- Statistics classes:

- CustomersStat: revenue, startdate, enddate
- SupplierStat: revenue, startdate, enddate
- SparePartStat: totalAmount, revenue, quantity, startdate, enddate
- ServiceStat: totalAmount, revenue, quantity, startdate, enddate

Step 4: Determine quantitative relationships between entities

- One Member can be a Staff or Customer.
- Staff specialized into ManagementStaff, WarehouseStaff, SaleStaff, TechnicalStaff.
- Customer owns one or more Vehicle(s).
- SaleStaff handles AppointmentOnline for Customer.
- WarehouseStaff manages ImportInvoice which contains ImportSparePart from Supplier.
- ReceivingSlip issued for ServiceSlip or SparePartSlip.
- SparePartSlip links SparePart to payment.
- ServiceSlip links Service to payment.
- Statistics tables (CustomersStat, SupplierStat, SparePartStat, ServiceStat) summarize transactional data.
- Address is a separate entity linked to Member and Supplier.

Step 5: Object relationships between entities

- Staff is abstract → specialized subclasses (ManagementStaff, WarehouseStaff, SaleStaff, TechnicalStaff).
- Customer is associated with Vehicle and AppointmentOnline.
- Supplier links to ImportSparePart and ImportInvoice.
- Service and SparePart link to their respective slip (ServiceSlip, SparePartSlip).
- ReceivingSlip aggregates payment details from both Service and SparePart slips.
- Statistics classes summarize data from main operational entities.
- Address reused by Member and Supplier entities.

Entity diagram

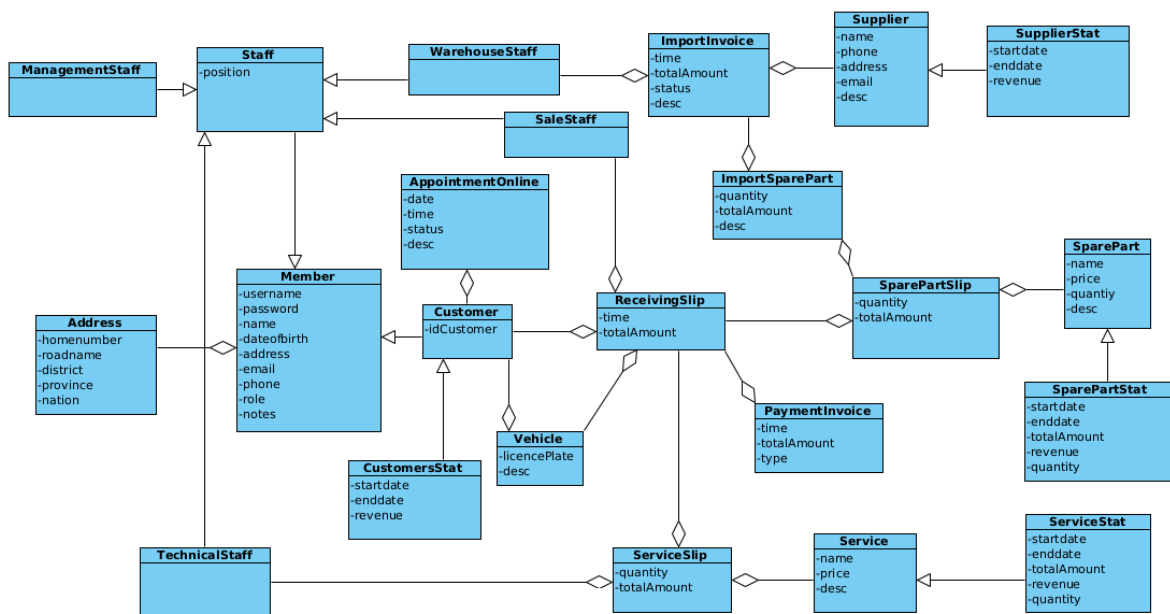


Figure 2.1. System analysis phase entity class diagram

3. Extract and draw the class diagram.

a. Analysis static of the Module “Customer Searches for Service/Parts Information”

Use Case 1: Customer search for service information

Initially, the main interface of the customer → propose class IFMainCustomer, with the following components:

- Select search for services: subSearchService (submit type)

Step 2: Service Search Interface → propose class IFSearchService, with the following components:

- inService: input box for entering the service name (input)
- outServiceList: list of found services (output)
- subSearch: button to perform the search (submit type)
- subViewDetail: button to view details of the selected service (submit type)
- subBack: button to return to the previous interface (submit type)

Step 3: Service Detail Interface → propose class IFDetailService, with the following components:

- outName: display the name of the selected service (output)
- outPrice: display the price or cost (output)

- outDescription: display the description/details of the service (output)
- subBack: button to return to the previous interface (submit type)

To obtain the Service list and detail information in the steps above, the system must process at the back end:

Find the Garage's services

- Input: Garage
- Output: Service list

Proposed method: getServiceList() and getServiceDetail() — assigned to class Service

Class Service includes:

- Attributes:
 - name
 - price
 - desc
- Methods:
 - getServiceList()
 - getServiceDetail()

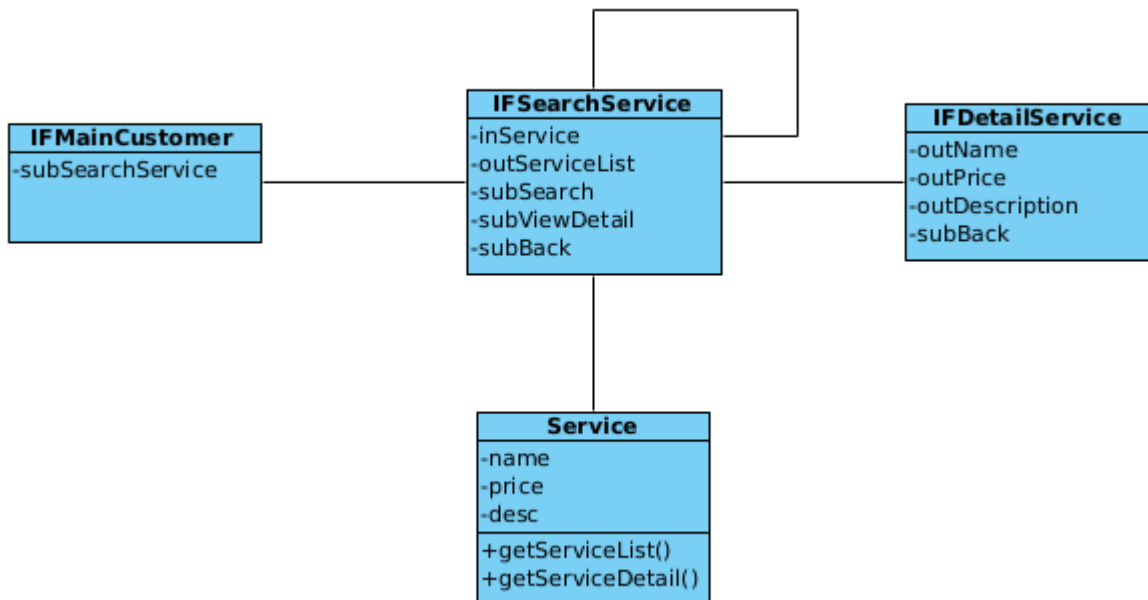


Figure 2.2. Class diagram for module search service

Use Case 2: Customer search for spare part information

Initially, the main interface of the customer → propose class IFMainCustomer, with the following components:

- Select search for spare parts: subSearchSparePart (submit type)

Step 2: Spare Part Search Interface → propose class IFSearchSparePart, with the following components:

- inSpareParts: input box for entering the spare part name (input)
- outSparePartList: list of found spare parts (output)
- subSearch: button to perform the search (submit type)
- subViewDetail: button to view details of the selected spare part (submit type)
- subBack: button to return to the previous interface (submit type)

Step 3: Spare Part Detail Interface → propose class IFDetailSparePart, with the following components:

- outItemName: display the name of the selected spare part (output)
- outPrice: display the price or cost (output)
- outQuantity: display available quantity (output)
- outDescription: display the description/details of the spare part (output)
- subBack: button to return to the previous interface (submit type)

To obtain the Spare Part list and detail information in the steps above, the system must process at the back end:

Find the Garage's spare parts

- Input: Garage
- Output: Spare part list

Proposed method: getSparePartList() and getSparePartDetail() — assigned to class SparePart

Class SparePart includes:

- Attributes:
 - name
 - price
 - quantity
 - desc
- Methods:
 - getSparePartList()
 - getSparePartDetail()

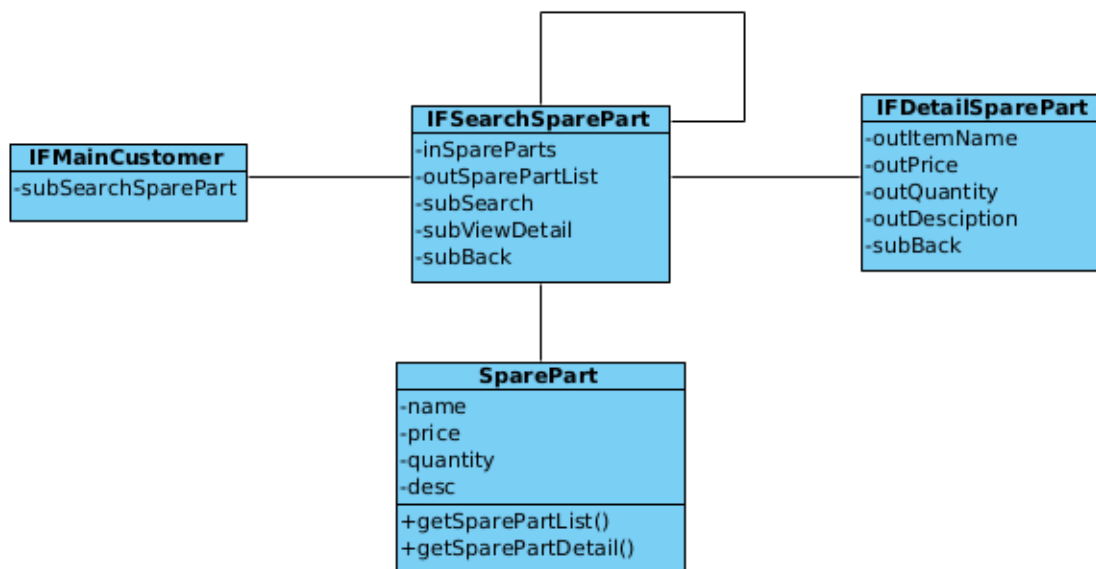


Figure 2.3. Class diagram for module search spare part

b. Analysis static of the Module “Management staff views customer statistics”

Initially, the main interface of the management staff → propose class IFMainManagementStaff, with the following components:

- Select menu “Statistical Reports”: subStatisticalReports (submit type)

Step 2: Select Customer Statistics by Revenue → propose class IFSelectStatistics, with the following components:

- subCustomerStatRevenue: choose “Customer Statistics by Revenue” (submit type)
- subBack: button to return to the main menu (submit type)

Methods:

- getSubBack()
- setSubBack(subBack): void

Step 3: View Customer Statistics Interface → propose class IFViewCustomerStatistics, with the following components:

- inStartDate: input start date (input)
- inEndDate: input end date (input)
- outCustomerList: list of customers and their revenue (output)
- subViewDetail: button to view a specific customer’s detail (submit type)
- subSearch: button to search for statistics (submit type)

- subBack: button to return to the previous interface (submit type)

Step 4: Displaying Customer's Invoices Interface → propose class IFViewCustomerInvoices, with the following components:

- outInvoiceList: list of invoices of the selected customer (output)
- subSelectInvoice: button to select a specific invoice (submit type)
- subBack: button to return to the previous interface (submit type)

Step 5: Displaying Detailed Invoice Interface → propose class IFInvoiceDetail, with the following components:

- outInvoiceDetail: display detailed information of the selected invoice (output)
- subBack: button to return to the invoice list (submit type)

Back-end processing to support these interfaces

1. Retrieve customer revenue statistics

- Input: startDate, endDate
- Output: list of customers with total revenue
- Proposed method: getCustomerRevenue() — assigned to class CustomerStat

Class CustomerStat includes:

- Attributes:
 - startDate
 - endDate
 - revenue
- Method:
 - getCustomerRevenue()

2. Retrieve invoices of a specific customer

- Input: idCustomer
- Output: list of invoices
- Proposed method: getPaymentInvoiceList() — assigned to class PaymentInvoice

Class PaymentInvoice includes:

- Attributes:
 - time
 - totalAmount
- Methods:
 - getPaymentInvoiceList()
 - getPaymentInvoiceDetail()

3. Retrieve detailed invoice information

- Input: invoice information (via PaymentInvoice)
- Output: full invoice details (services/parts used, quantity, total, etc.)
- Proposed method: `getPaymentInvoiceDetail()` — assigned to class `PaymentInvoice`

Supporting classes:

- Customer:
 - `idCustomer`
- Vehicle:
 - `licensePlate`, `desc`
- ReceivingSlip:
 - `time`, `totalAmount`

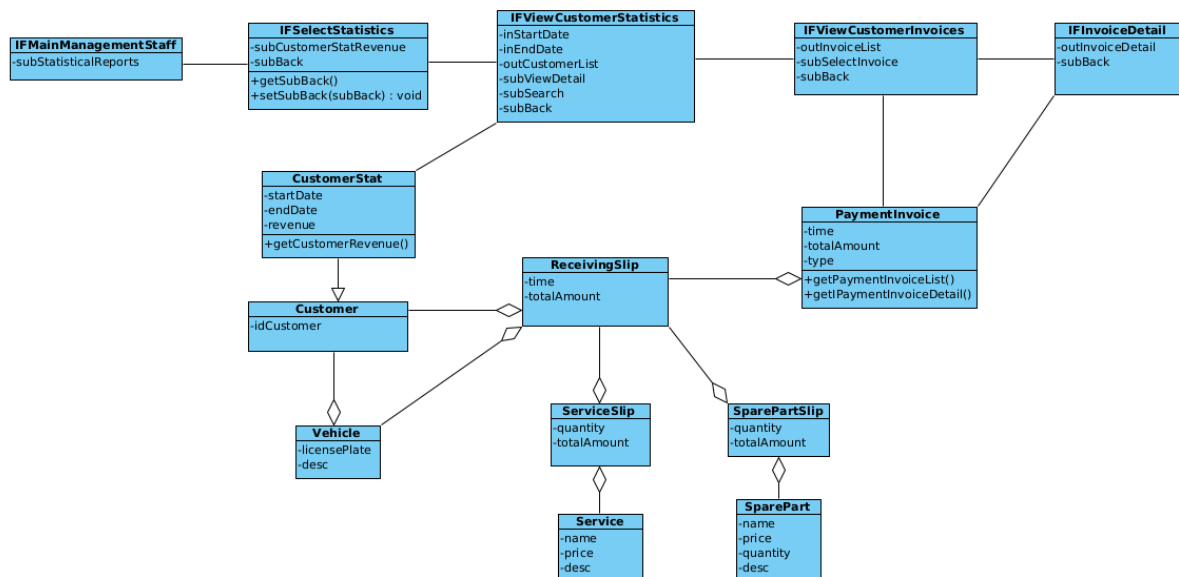


Figure 2.4. Class diagram for module view customer statistics

4. State diagram

a. Module “Customer Searches for Service/Spare Parts Information”

State Diagram Description for the Module “Customer Searches for Service/Parts Information”

- From the Customer’s Main Interface (Displaying customer’s main interface):
 - If the customer selects [Select Find Service Information], the system moves to the Interface Service Search (Displaying Interface Service List).

- If the customer selects [Select Find Spare Part Information], the system moves to the Interface Spare Parts Search (Displaying Interface Spare Part List).
- At the Interface Service Search (Displaying Interface Service List):
 - The customer [Enter keyword and click Search] → the system retrieves and displays the list of matching services on the same interface.
 - If the customer selects [Select View Detail] on a service item, the system moves to Interface Service Detail (Displaying Interface Service Detail).
 - If the customer clicks [Select Back Button], the system returns to the Customer's Main Interface.
- At the Interface Spare Parts Search (Displaying Interface Spare Part List):
 - The customer [Enter keyword and click Search] → the system retrieves and displays the list of matching spare parts on the same interface.
 - If the customer selects [Select View Detail] on a spare part item, the system moves to Interface Spare Parts Detail (Displaying Interface Spare Part Detail).
 - If the customer clicks [Select Back Button], the system returns to the Customer's Main Interface.
- At the Interface Service Detail (Displaying Interface Service Detail):
 - The system displays full details of the selected service.
 - If the customer clicks [Select Back Button], the system returns to the Interface Service Search (Displaying Interface Service List).
 - If the customer clicks [Select Home], the system returns to the Customer's Main Interface.
 - The flow ends at End state.
- At the Interface Spare Parts Detail (Displaying Interface Spare Part Detail):
 - The system displays full details of the selected spare part.
 - If the customer clicks [Select Back Button], the system returns to the Interface Spare Part Search (Displaying Interface Spare Part List).
 - If the customer clicks [Select Home], the system returns to the Customer's Main Interface.
 - The flow ends at End state.

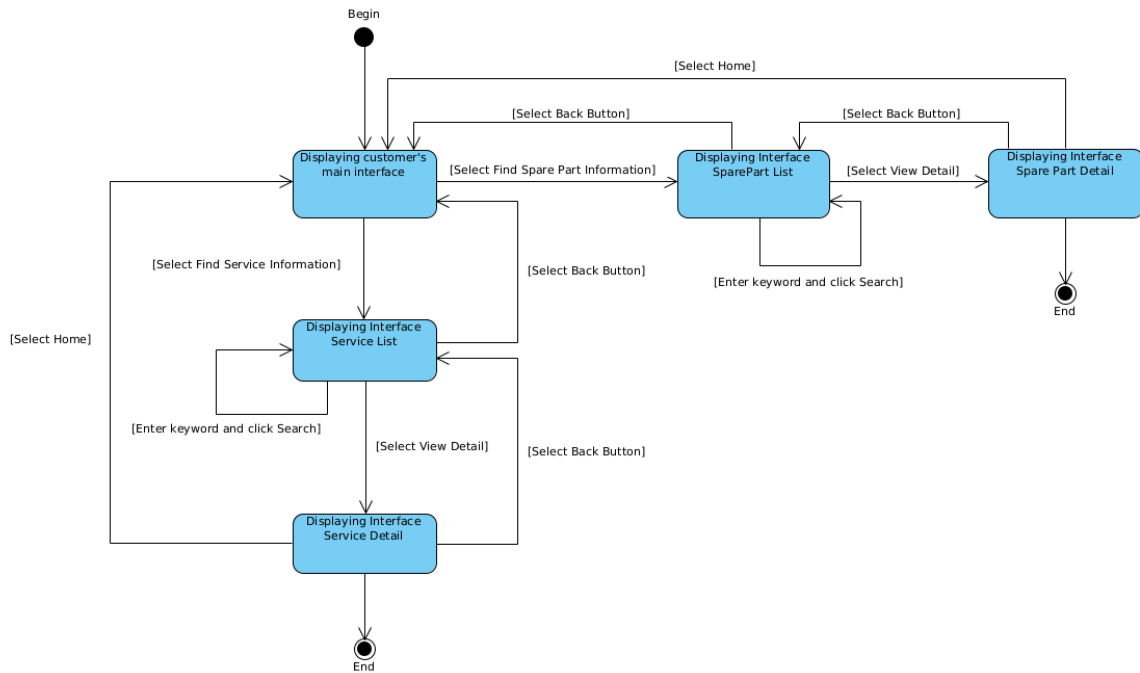


Figure 2.5. State diagram for module search service/spare parts information

b. Module “Management staff views customer statistics”

State Diagram Description for the Module “Management Staff Views Customer Statistics”

- From the Management Staff’s Main Interface (Displaying Management Staff’s Main Interface):
 - If the management staff selects [Select View Statistical Report], the system moves to the Interface Select Customer Statistics by Revenue (Displaying Interface Select Customer Statistics by Revenue).
- At the Interface Select Statistics (Displaying Interface Select Statistics):
 - If the management staff selects [Select Customer Revenue Statistics], the system moves to the Interface View Customer Statistics (Displaying Interface View Customer Statistics).
 - If the management staff clicks [Select Back Button], the system returns to the Management Staff’s Main Interface.
- At the Interface View Customer Statistics (Displaying Interface View Customer Statistics):
 - The management staff [Enter date range and click Generate button] → the system retrieves the list of customers with their total revenue and displays it in the Interface View Customer Statistics (Displaying Interface View Customer Statistics).
 - If the management staff [Selects a Specific Customer], the system retrieves that customer’s invoices and moves to Interface View Customer Invoices (Displaying Interface View Customer Invoices).

- If the management staff clicks [Select Back Button], the system returns to the Interface Select Statistics.
- At the Interface View Customer Invoices (Displaying Interface View Customer Invoices):
 - The system displays a list of invoices for the selected customer.
 - If the management staff [Selects a Specific Invoice], the system retrieves the full invoice details and moves to Interface Invoice Detail (Displaying Interface Invoice Detail).
 - If the management staff clicks [Select Back Button], the system returns to the Interface View Customer Statistics.
- At the Interface Invoice Detail (Displaying Interface Invoice Detail):
 - The system displays full invoice details, including services or parts used, quantity, unit price, total amount, etc.
 - If the management staff clicks [Select Back Button], the system returns to the Interface View Customer Invoices.
 - If the management staff clicks [Home], the system returns to the Management Staff's Main Interface.
 - The flow ends at the End state.

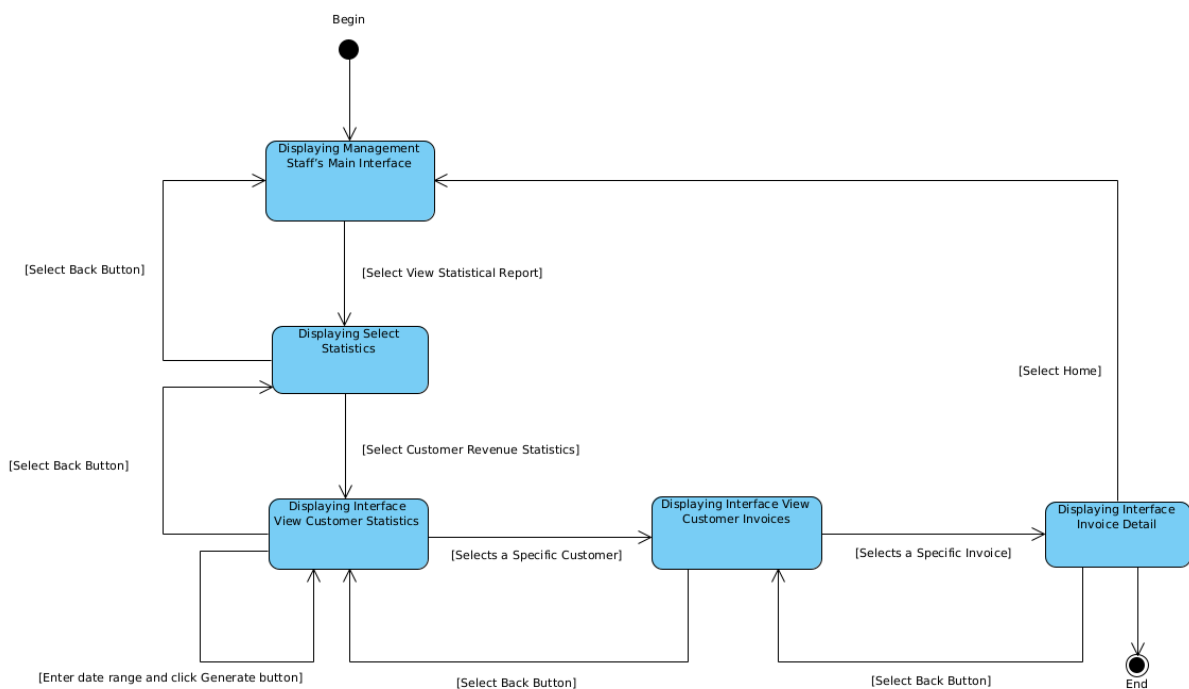


Figure 2.6. State diagram for module view customer statistics

5. Draw the communication diagram for the module.

a. Module “Customer Searches for Service/Spare Parts Information”

Scenario v.2 for Module “Customer Searches for Service Information”

1. At the customer's main interface after login, the Customer clicks the button to search for services.
2. Class IFMainCustomer calls class IFSearchService.
3. Class IFSearchService displays the search interface for the customer.
4. The Customer enters a keyword into the input field.
5. Class IFSearchService calls class Service to request the list of services that match the entered keyword.
6. Class Service executes the method getServiceList() to retrieve the matching services from the system database.
7. Class Service returns the results to IFSearchService.
8. Class IFSearchService displays the list of matching services to the Customer.
9. The Customer clicks on one specific service from the displayed list.
10. Class IFSearchService calls class IFDetailService.
11. Class IFDetailService calls class Service to request details of the selected service.
12. Class Service executes the method getServiceDetail() to retrieve full information (name, description, price) of the selected service.
13. Class Service returns the detailed information to IFDetailService.
14. Class IFDetailService displays the selected service's name, description, and price to the Customer.

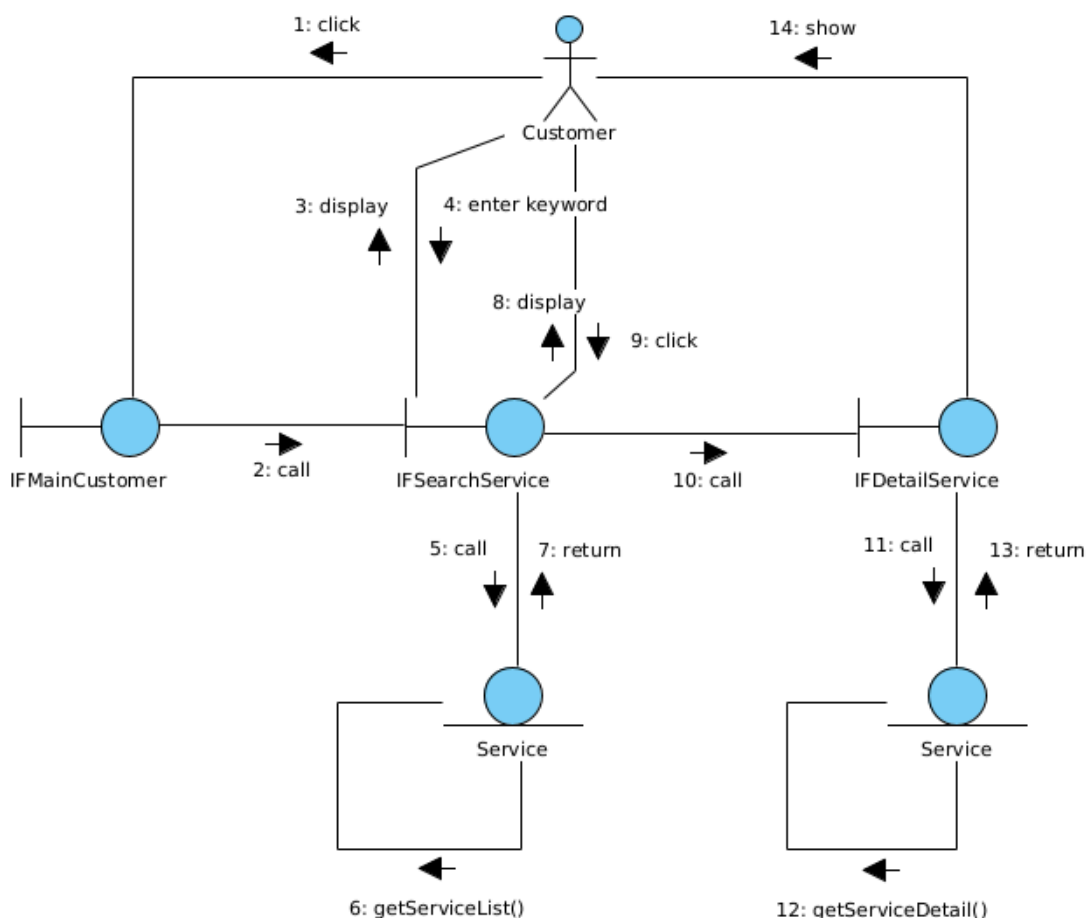


Figure 2.7. Communication diagram for module search service information

Scenario v.2 for Module “Customer Searches for Spare Parts Information”

1. At the customer's main interface after login, the Customer clicks the button to search for spare parts.
2. Class IFMainCustomer calls class IFSearchSparePart.
3. Class IFSearchSparePart displays the search interface for the customer.
4. The Customer enters a keyword into the input field.
5. Class IFSearchSparePart calls class SparePart to request a list of spare parts that match the entered keyword.
6. Class SparePart executes the method getSparePartList() to retrieve the list of spare parts from the system database.
7. Class SparePart returns the results to IFSearchSparePart.
8. Class IFSearchSparePart displays the list of matching spare parts to the Customer.
9. The Customer clicks on one specific spare part from the displayed list.
10. Class IFSearchSparePart calls class IFDetailSparePart.
11. Class IFDetailSparePart calls class SparePart to request details of the selected spare part.
12. Class SparePart executes the method getSparePartDetail() to retrieve full information (name, description, quantity, price) of the selected spare part.
13. Class SparePart returns the detailed information to IFDetailSparePart.
14. Class IFDetailSparePart displays the selected spare part's name, description, quantity, and price to the Customer.

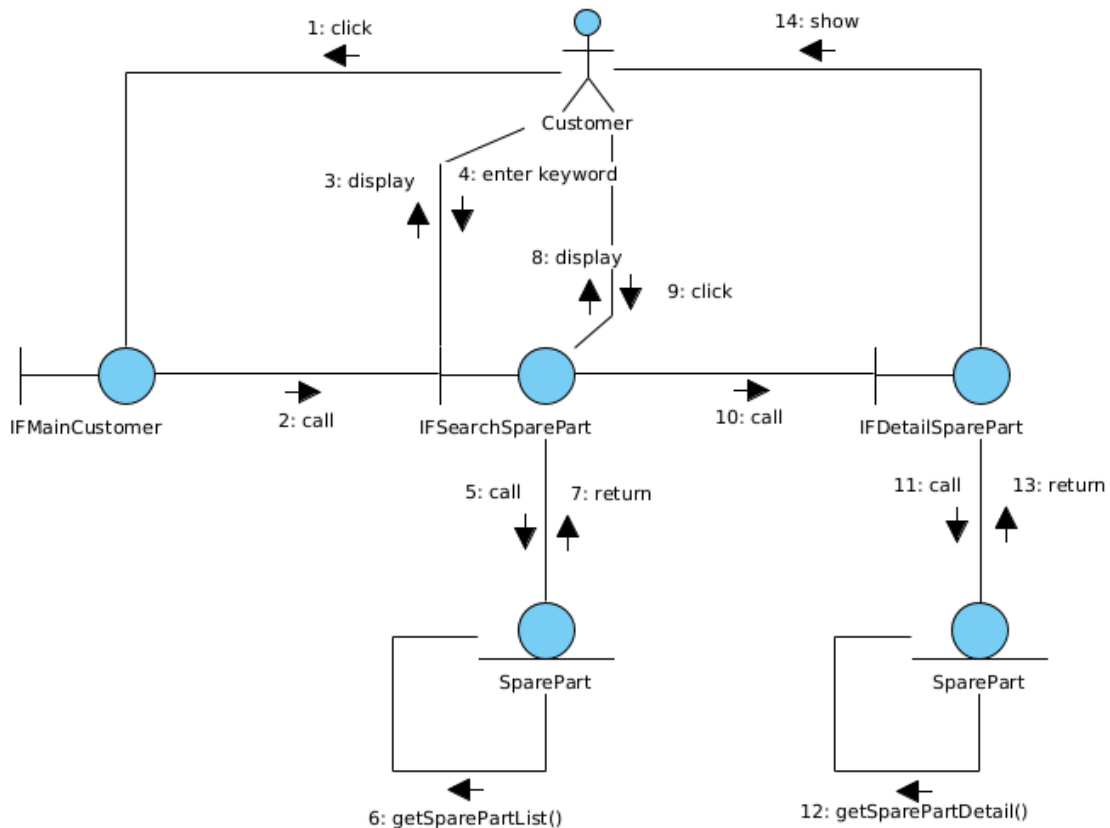


Figure 2.8. Communication diagram for module search spare parts information

b. Module “Management staff views customer statistics”

Scenario v.2 for Module “Management Staff Views Customer Statistics”

1. At the main interface of the management staff after login, the Management Staff clicks the “Statistical Reports” menu.
2. Class IFMainManagementStaff calls class IFSelectStatistics to open the statistics selection interface.
3. Class IFSelectStatistics displays the interface that allows the management staff to choose the type of report.
4. The Management Staff clicks to select the Customer Revenue Statistics option.
5. Class IFSelectStatistics calls class IFViewCustomerStatistics.
6. Class IFViewCustomerStatistics displays the interface for viewing customer statistics by revenue.
7. The Management Staff enters the date range (start date and end date).
8. Class IFViewCustomerStatistics calls class CustomerStat to request the revenue data of customers within the selected period.
9. Class CustomerStat executes the method `getCustomerRevenue()` to retrieve customer revenue data from the database.

10. Class CustomerStat returns the results to IFViewCustomerStatistics.
11. Class IFViewCustomerStatistics displays the list of customers and their corresponding revenue.
12. The Management Staff clicks on a specific customer to view all invoices related to that customer.
13. Class IFViewCustomerStatistics calls class PaymentInvoice to request the invoices of the selected customer.
14. Class PaymentInvoice executes the method getPaymentInvoiceList() to retrieve all invoices for that customer from the database.
15. Class PaymentInvoice returns the list of invoices to IFViewCustomerStatistics.
16. Class IFViewCustomerStatistics calls class IFViewCustomerInvoices.
17. Class IFViewCustomerInvoices displays the list of invoices to the Management Staff.
18. The Management Staff clicks on a specific invoice to view its details.
19. Class IFViewCustomerInvoices calls class PaymentInvoice to request detailed information for the selected invoice.
20. Class PaymentInvoice executes the method getPaymentInvoiceDetail() to retrieve all invoice details (services/parts, quantity, total cost, etc.) from the database.
21. Class PaymentInvoice returns the detailed invoice information to IFViewCustomerInvoices.
22. Class IFViewCustomerInvoices calls class IFInvoiceDetail.
23. Class IFInvoiceDetail displays all details of the selected invoice, including services, spare parts, quantities, and total cost, to the Management Staff.

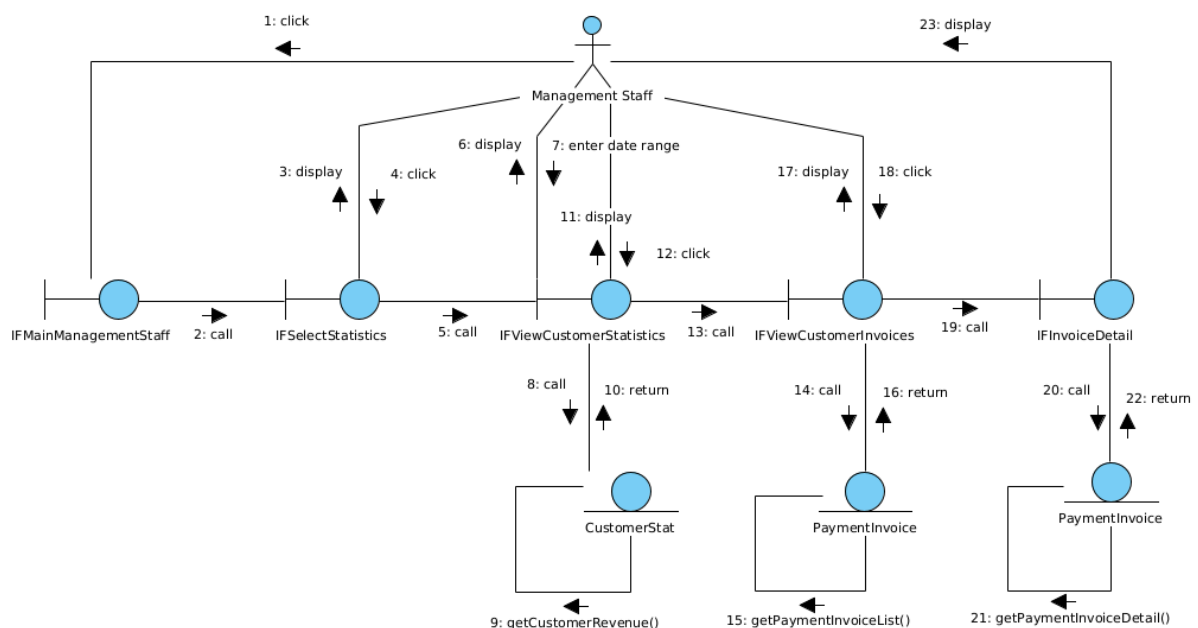


Figure 2.9. Communication diagram for module search service information

6. Sequence diagram

a. Module “Customer Searches for Service/Spare Parts Information”

Usecase 1: Customer search for service information.

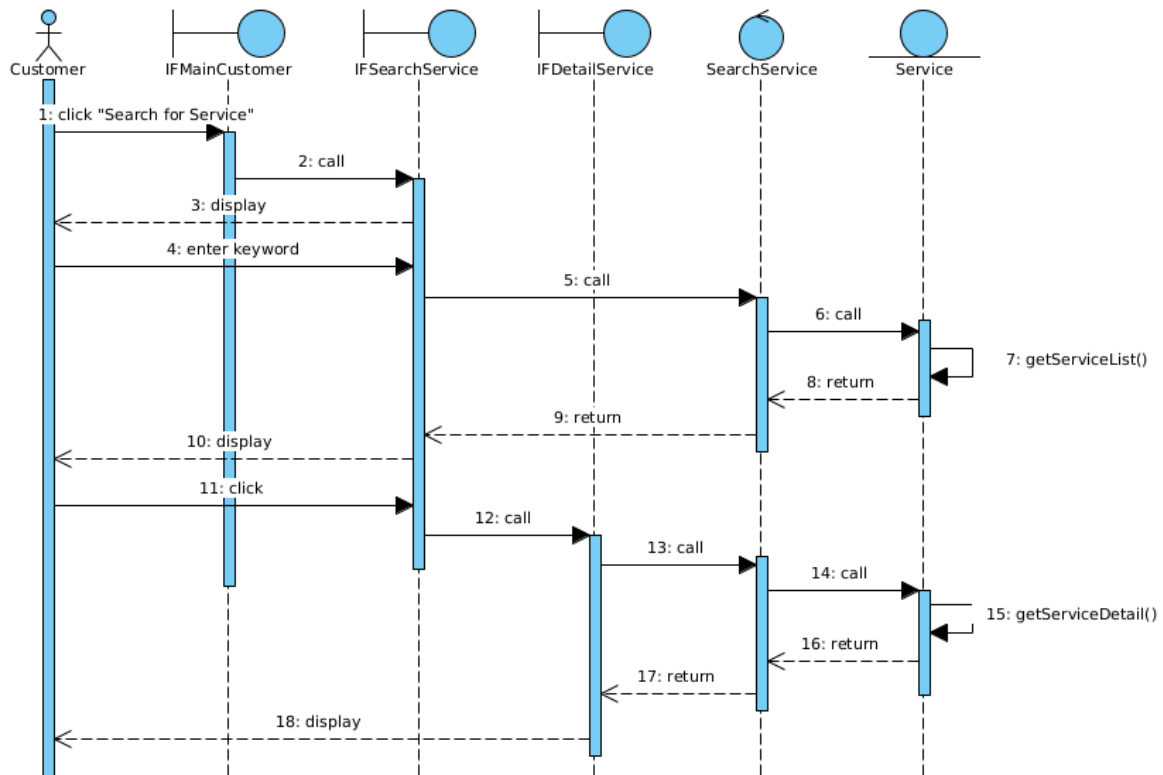


Figure 2.10. Sequence diagram for module search service information

Usecase 2: Customer search for spare parts information.

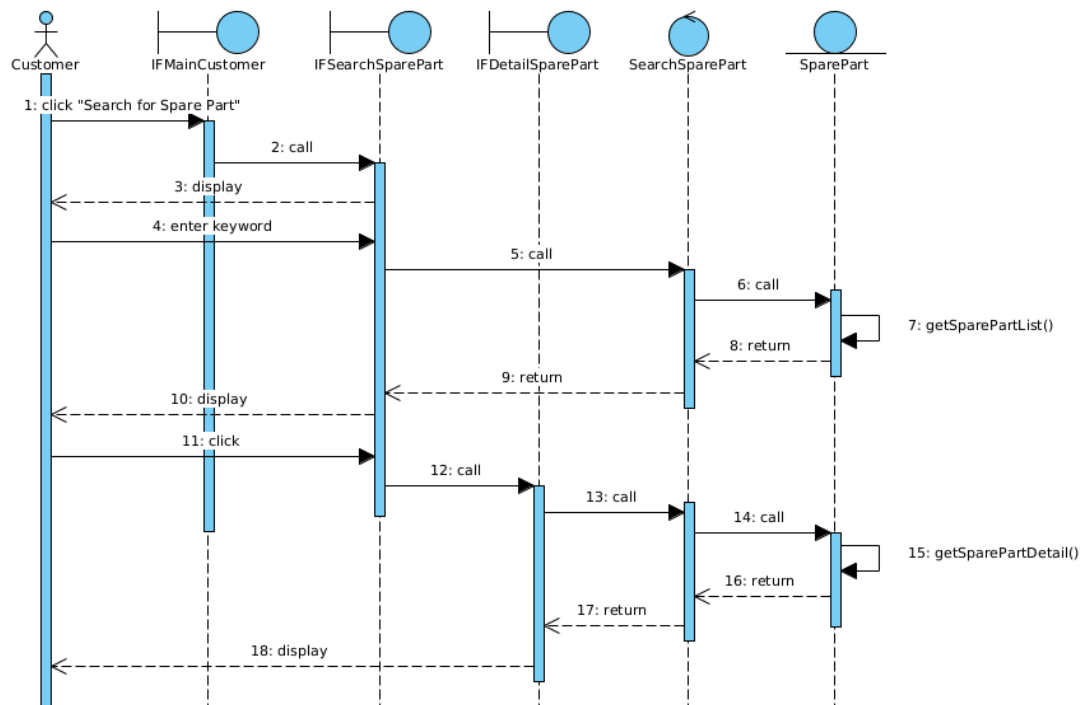


Figure 2.11. Sequence diagram for module search spare part information

b. Module “Management staff views customer statistics”

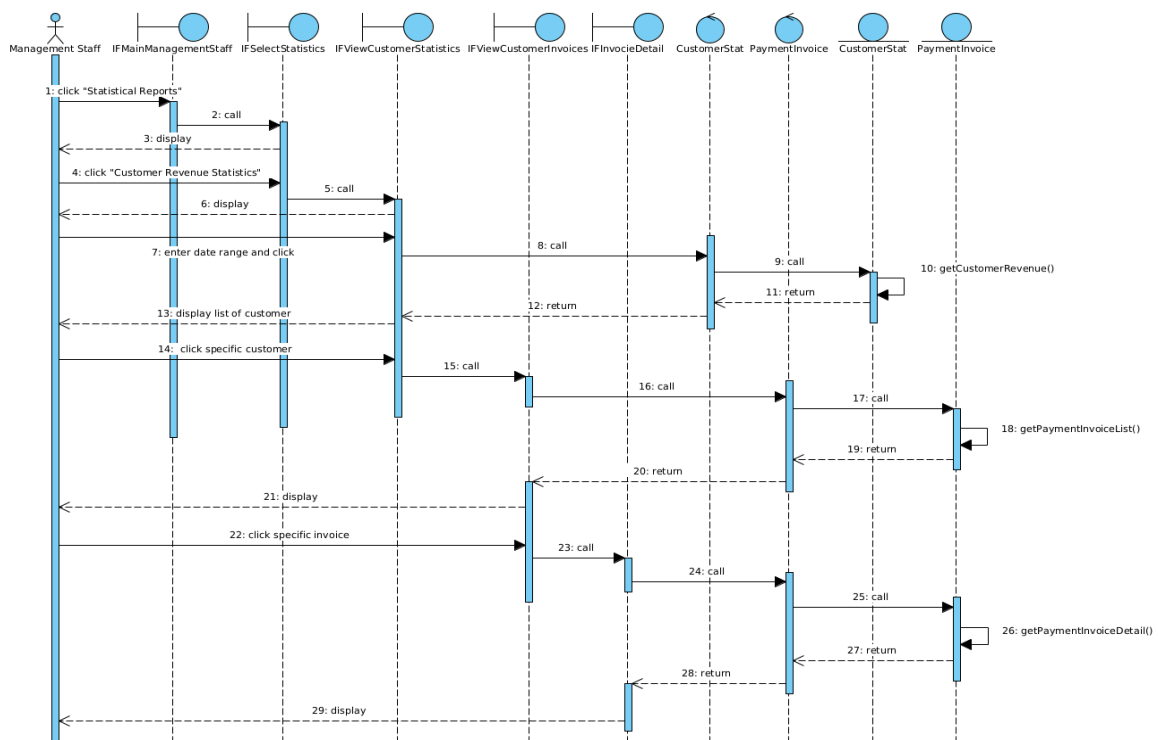


Figure 2.12. Sequence diagram for module search service information

III. Design

1. Entity class diagram

Step 1

The following classes are supplemented with an id attribute, except for the classes Staff, Member, and Address (since they already have one):

- Customer
- Vehicle
- AppointmentOnline
- ReceivingSlip
- PaymentInvoice
- ImportInvoice
- Supplier
- ImportSparePart
- SparePartSlip
- SparePart
- ServiceSlip
- Service
- SupplierStat
- SparePartStat
- ServiceStat
- CustomerStat

Step 2

The attributes of all classes are supplemented with appropriate Java data types:

- **Staff:**
 - position: String
- **Member:**
 - id: String
 - username: String
 - password: String
 - name: String
 - dateofbirth: java.util.Date
 - address: Address
 - email: String
 - phone: String
 - role: String
 - notes: String
- **Address:**
 - id: String

- homenumber: String
- roadname: String
- district: String
- province: String
- nation: String
- **Customer:**
 - idCustomer: String
- **Vehicle:**
 - id: String
 - customer: Customer
 - licensePlate: String
 - desc: String
- **AppointmentOnline:**
 - id: String
 - customer: Customer
 - date: java.util.Date
 - status: String
 - desc: String
- **ReceivingSlip:**
 - id: String
 - customer: Customer
 - vehicle: Vehicle
 - time: java.time.LocalDateTime
 - totalAmount: BigDecimal
 - saleStaff: Staff
- **PaymentInvoice:**
 - id: String
 - receivingSlip: ReceivingSlip
 - time: java.time.LocalDateTime
 - totalAmount: BigDecimal
 - type: String
- **ImportInvoice:**
 - id: String
 - time: java.time.LocalDateTime
 - totalAmount: BigDecimal
 - desc: String
 - warehouseStaff: Staff
 - supplier: Supplier
- **Supplier:**
 - id: String
 - name: String
 - phone: String
 - address: String
 - email: String

- desc: String
- **ImportSparePart:**
 - id: String
 - importInvoice: ImportInvoice
 - sparepart: SparePart
 - quantity: int
 - totalAmount: BigDecimal
 - desc: String
- **SparePartSlip:**
 - id: String
 - sparepart: SparePart
 - receivingSlip: ReceivingSlip
 - quantity: int
 - totalAmount: BigDecimal
- **SparePart:**
 - id: String
 - name: String
 - price: BigDecimal
 - quantity: int
 - desc: String
- **ServiceSlip:**
 - id: String
 - service: Service
 - receivingSlip: ReceivingSlip
 - quantity: int
 - totalAmount: BigDecimal
 - technicalStaff: Staff
- **Service:**
 - id: String
 - name: String
 - price: BigDecimal
 - desc: String
- **SupplierStat:**
 - revenue: BigDecimal
 - startdate: java.util.Date
 - enddate: java.util.Date
- **SparePartStat:**
 - totalAmount: BigDecimal
 - revenue: BigDecimal
 - quantity: int
 - startdate: java.util.Date
 - enddate: java.util.Date
- **ServiceStat:**
 - totalAmount: BigDecimal

- revenue: BigDecimal
- quantity: int
- startdate: java.util.Date
- enddate: java.util.Date
- **CustomerStat:**
 - revenue: BigDecimal
 - number: int
 - startdate: java.util.Date
 - enddate: java.util.Date

Step 3

The relationships are transformed into composition or aggregation attributes as follows:

- Customer – Vehicle → Vehicle contains customer: Customer
- Customer – ReceivingSlip → ReceivingSlip contains customer: Customer
- ReceivingSlip – PaymentInvoice → PaymentInvoice contains receivingSlip: ReceivingSlip
- ImportInvoice – Supplier → ImportInvoice contains supplier: Supplier
- ImportInvoice – ImportSparePart → ImportSparePart contains importInvoice: ImportInvoice
- ReceivingSlip – SparePartSlip → SparePartSlip contains receivingSlip: ReceivingSlip
- SparePartSlip – SparePart → SparePartSlip contains sparepart: SparePart
- ReceivingSlip – ServiceSlip → ServiceSlip contains receivingSlip: ReceivingSlip
- ServiceSlip – Service → ServiceSlip contains service: Service
- ServiceSlip – Staff → ServiceSlip contains technicalStaff: Staff
- ReceivingSlip – Staff → ReceivingSlip contains saleStaff: Staff
- ImportInvoice – Staff → ImportInvoice contains warehouseStaff: Staff

Step 4

Additional component attributes for composition and aggregation relationships are supplemented to complete the Entity Class Diagram in the Design Phase, as shown in the diagram you provided.

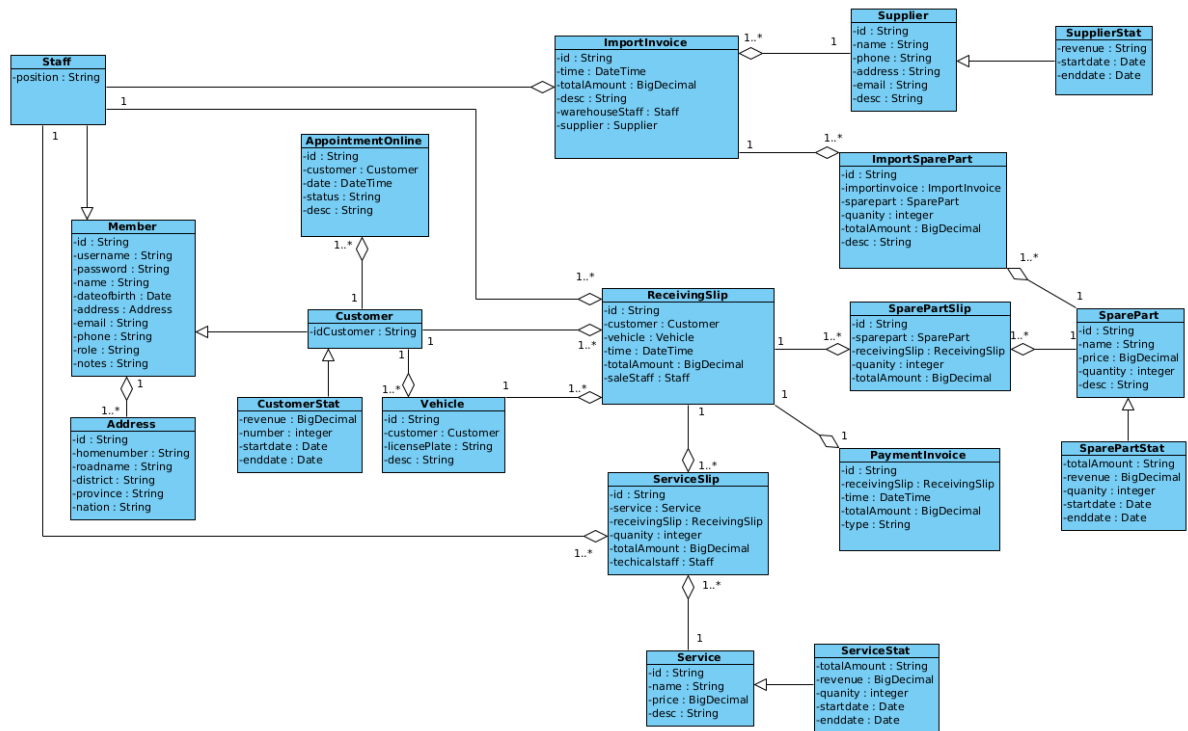


Figure 3.1. Entity class design diagram

2. Database design

Note: desc in class diagram => **description** in database

Step 1: Table Mapping

Each entity class is mapped to a corresponding table as follows:

- Staff → **tblStaff**
- Member → **tblMember**
- Address → **tblAddress**
- Customer → **tblCustomer**
- Vehicle → **tblVehicle**
- AppointmentOnline → **tblAppointmentOnline**
- ReceivingSlip → **tblReceivingSlip**
- PaymentInvoice → **tblPaymentInvoice**
- ImportInvoice → **tblImportInvoice**
- Supplier → **tblSupplier**
- ImportSparePart → **tblImportSparePart**
- SparePartSlip → **tblSparePartSlip**
- SparePart → **tblSparePart**
- ServiceSlip → **tblServiceSlip**
- Service → **tblService**

Step 2: Attribute Conversion

The attributes of the classes are converted into table columns with correct data types and constraints.

- **tblStaff:**
 - tblMemberid **varchar(255)** [PK, FK → tblMember.id]
 - position **varchar(255)** [N]
- **tblMember:**
 - id **varchar(255)** [PK, U]
 - username **varchar(255)** [U]
 - password **varchar(255)**
 - name **varchar(255)**
 - dateofbirth **date**
 - email **varchar(255)**
 - phone **varchar(255)**
 - role **varchar(255)**
 - note **varchar(255)**
 - tblAddressid **varchar(255)** [FK → tblAddress.id]
- **tblAddress:**
 - id **varchar(255)** [PK, U]
 - homenummer **varchar(255)**
 - roadname **varchar(255)**
 - district **varchar(255)**
 - province **varchar(255)**
 - nation **varchar(255)**
- **tblCustomer:**
 - idCustomer **varchar(255)** [PK]
 - tblMemberid **varchar(255)** [FK → tblMember.id] [N]
- **tblVehicle:**
 - id **varchar(255)** [PK, U]
 - licensePlate **varchar(255)**
 - description **varchar(255)** [N]
 - tblCustomertblMemberid **varchar(255)** [FK → tblCustomer.tblMemberid]
- **tblAppointmentOnline:**
 - id **varchar(255)** [PK, U]
 - time **timestamp**
 - status **varchar(255)**
 - description **varchar(255)** [N]
 - tblCustomertblMemberid **varchar(255)** [FK → tblCustomer.tblMemberid] [N]
- **tblReceivingSlip:**
 - id **varchar(255)** [PK, U]

- time **timestamp**
- totalAmount **decimal(15,2)**
- description **varchar(255)** [N]
- tblVehicleid **varchar(255)** [FK → tblVehicle.id]
- tblCustomerTblMemberid **varchar(255)** [FK → tblCustomer.tblMemberid]
- tblStaffTblMemberid **varchar(255)** [FK → tblStaff.tblMemberid]
- **tblPaymentInvoice:**
 - id **varchar(255)** [PK, U]
 - totalAmount **decimal(15,2)**
 - time **timestamp**
 - type **varchar(255)**
 - tblReceivingSlipid **varchar(255)** [FK → tblReceivingSlip.id]
- **tblImportInvoice:**
 - id **varchar(255)** [PK, U]
 - time **timestamp**
 - totalAmount **decimal(15,2)**
 - status **varchar(255)**
 - description **varchar(255)** [N]
 - Supplierid **varchar(255)** [FK → tblSupplier.id]
 - tblStaffTblMemberid **varchar(255)** [FK → tblStaff.tblMemberid]
- **tblSupplier:**
 - id **varchar(255)** [PK, U]
 - name **varchar(255)**
 - phone **varchar(255)**
 - address **varchar(255)**
 - email **varchar(255)** [N]
 - description **varchar(255)** [N]
- **tblImportSparePart:**
 - id **varchar(255)** [PK, U]
 - quantity **integer(10)**
 - totalAmount **decimal(15,2)**
 - description **varchar(255)** [N]
 - SparePartid **varchar(255)** [FK → tblSparePart.id]
 - ImportInvoiceid **varchar(255)** [FK → tblImportInvoice.id]
- **tblSparePartSlip:**
 - id **varchar(255)** [PK, U]
 - quantity **integer(10)**
 - totalAmount **decimal(15,2)**
 - SparePartid **varchar(255)** [FK → tblSparePart.id]
 - tblReceivingSlipid **varchar(255)** [FK → tblReceivingSlip.id]
- **tblSparePart:**
 - id **varchar(255)** [PK, U]
 - name **varchar(255)**

- price **decimal(15,2)**
- quantity **integer(10)**
- description **varchar(255)** [N]
- **tblServiceSlip:**
 - id **varchar(255)** [PK, U]
 - quantity **integer(10)**
 - totalAmount **decimal(15,2)**
 - tblServiceid **varchar(255)** [FK → tblService.id]
 - tblStafftblMemberid **varchar(255)** [FK → tblStaff.tblMemberid]
 - tblReceivingSlipid **varchar(255)** [FK → tblReceivingSlip.id]
- **tblService:**
 - id **varchar(255)** [PK, U]
 - name **varchar(255)**
 - price **decimal(15,2)**
 - description **varchar(255)** [N]

Step 3: Relationship Conversion

Relationships among the tables are implemented using foreign keys.

- **1 tblMember – n tblStaff:** tblStaff.tblMemberid → tblMember.id
- **1 tblMember – n tblCustomer:** tblCustomer.tblMemberid → tblMember.id
- **1 tblCustomer – n tblVehicle:** tblVehicle.tblCustomertblMemberid → tblCustomer.tblMemberid
- **1 tblCustomer – n tblAppointmentOnline:**
tblAppointmentOnline.tblCustomertblMemberid → tblCustomer.tblMemberid
- **1 tblCustomer – n tblReceivingSlip:**
tblReceivingSlip.tblCustomertblMemberid → tblCustomer.tblMemberid
- **1 tblVehicle – n tblReceivingSlip:** tblReceivingSlip.tblVehicleid → tblVehicle.id
- **1 tblStaff – n tblReceivingSlip:** tblReceivingSlip.tblStafftblMemberid → tblStaff.tblMemberid
- **1 tblReceivingSlip – 1 tblPaymentInvoice:**
tblPaymentInvoice.tblReceivingSlipid → tblReceivingSlip.id
- **1 tblSupplier – n tblImportInvoice:** tblImportInvoice.Supplierid → tblSupplier.id
- **1 tblStaff – n tblImportInvoice:** tblImportInvoice.tblStafftblMemberid → tblStaff.tblMemberid
- **1 tblImportInvoice – n tblImportSparePart:**
tblImportSparePart.ImportInvoiceid → tblImportInvoice.id
- **1 tblSparePart – n tblImportSparePart:** tblImportSparePart.SparePartid → tblSparePart.id
- **1 tblReceivingSlip – n tblSparePartSlip:** tblSparePartSlip.tblReceivingSlipid → tblReceivingSlip.id

- **1 tblSparePart – n tblSparePartSlip:** tblSparePartSlip.SparePartid → tblSparePart.id
- **1 tblReceivingSlip – n tblServiceSlip:** tblServiceSlip.tblReceivingSlipid → tblReceivingSlip.id
- **1 tblService – n tblServiceSlip:** tblServiceSlip.tblServiceid → tblService.id
- **1 tblStaff – n tblServiceSlip:** tblServiceSlip.tblStafftblMemberid → tblStaff.tblMemberid

Step 4: Key Conversion

- **Primary Keys (PK):** id or idXXX in each main table (e.g., tblMember.id, tblCustomer.idCustomer, tblVehicle.id, etc.)
- **Foreign Keys (FK):** As defined in Step 3
- **Unique Constraints (U):**
 - tblMember.id, tblMember.username
 - tblAddress.id
 - All id fields in entity tables (tblVehicle.id, tblService.id, etc.)

Step 5: Derived and Removed Attributes

- Derived attributes (e.g., totalAmount in tblReceivingSlip, tblPaymentInvoice) are calculated dynamically and not stored physically if possible.
- Statistic-related tables (tblSupplierStat, tblSparePartStat, tblServiceStat, tblCustomerStat) are **excluded** from the physical model to ensure normalization and reduce redundancy.

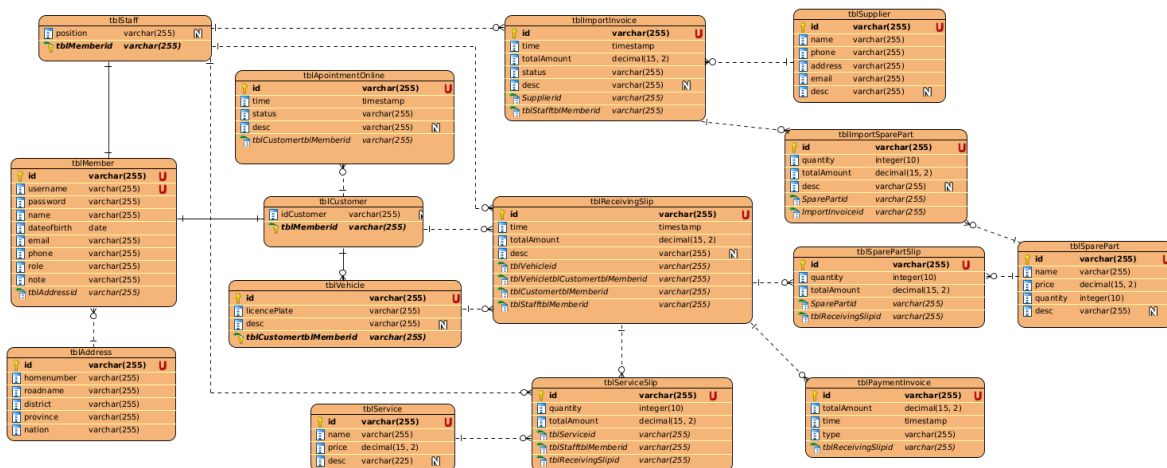
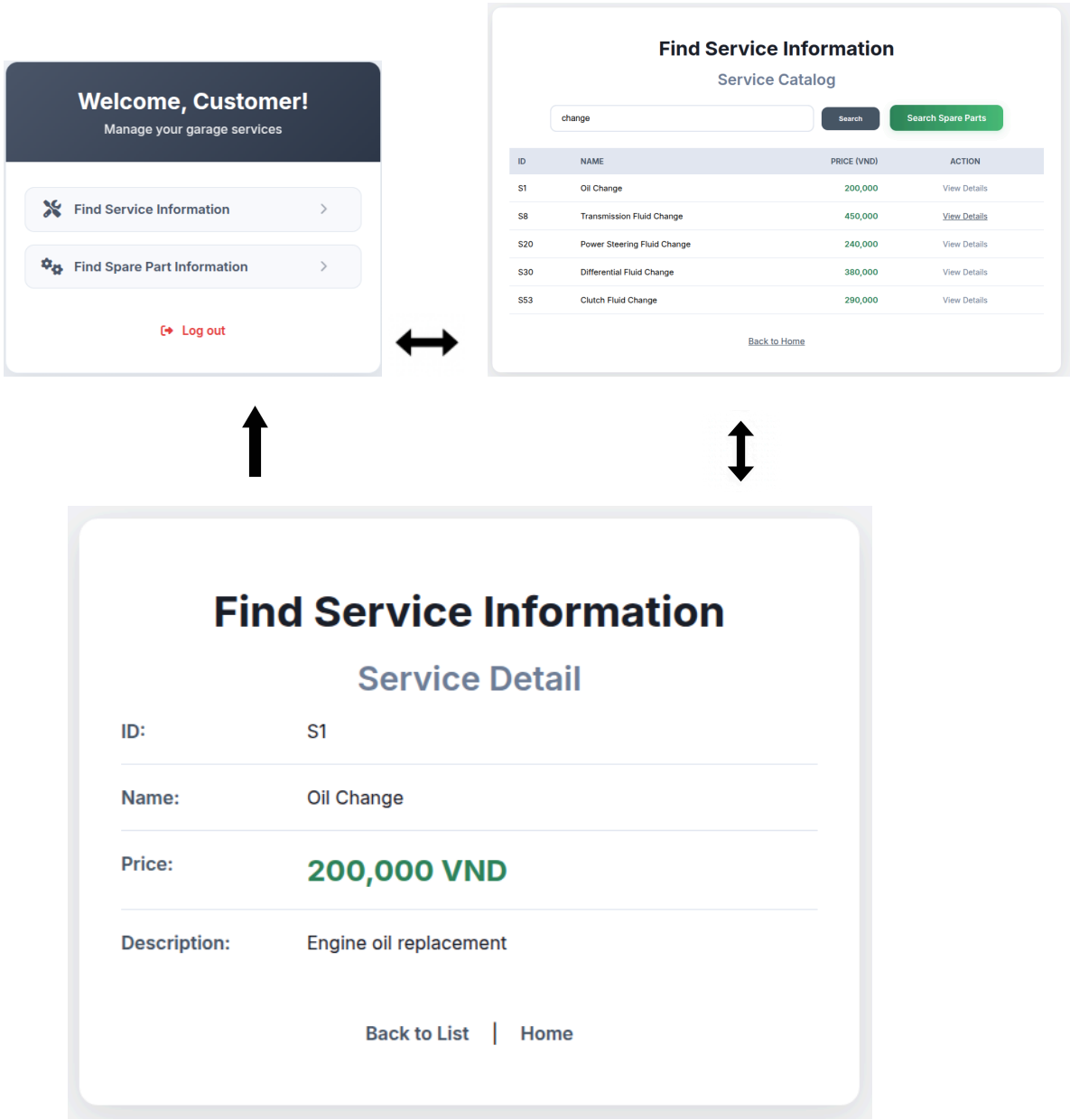


Figure 3.2. Database design diagram

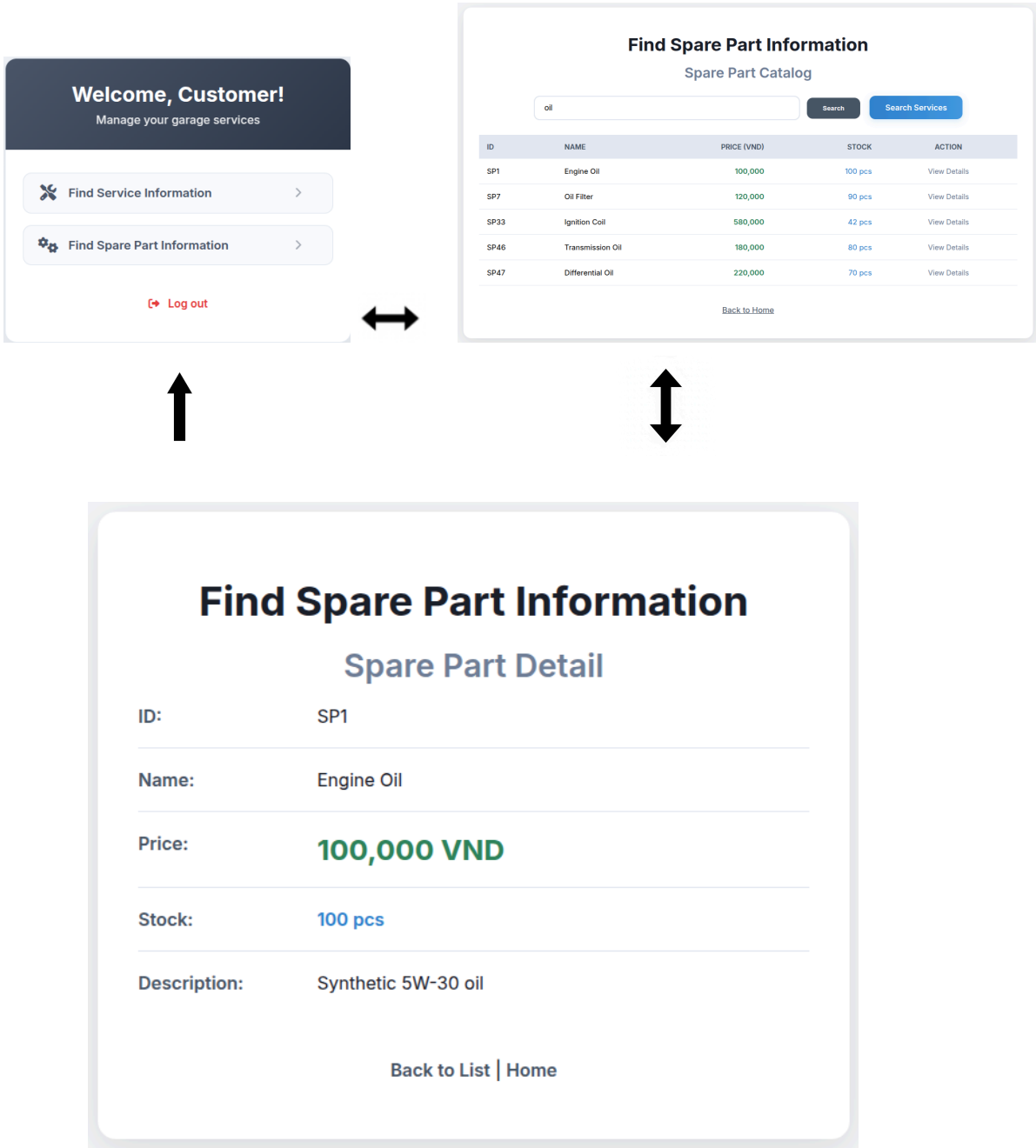
3. Interface

a. Module 1: Customer searches for service/spare parts information

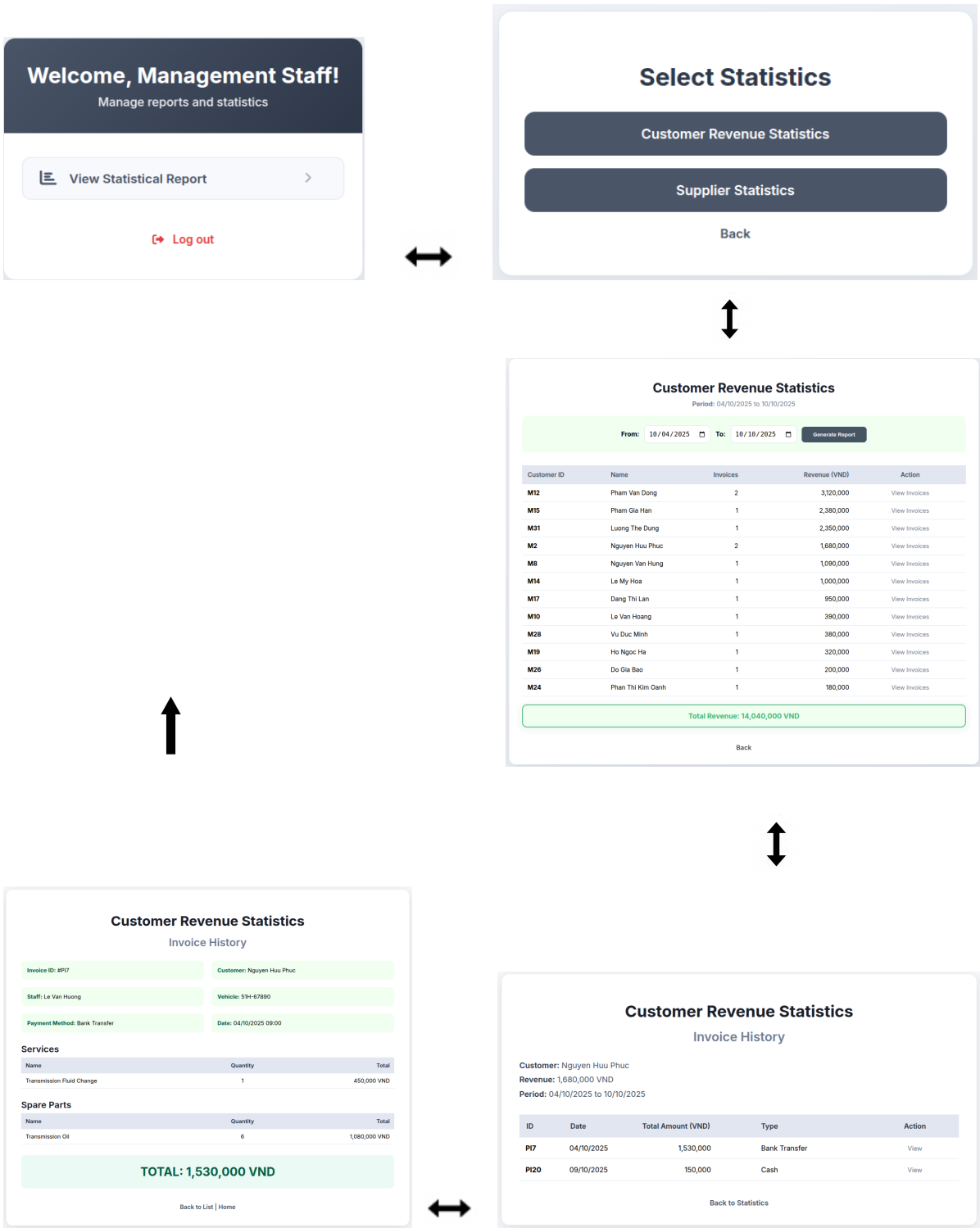
Usecase 1: Customer searches for service information



Usecase 2: Customer searches for spare parts information



b. Module 2: Management staff views customer statistics



4. Class diagram for each module

a. Module 1: Customer searches for service/spare parts information

Usecase 1: Customer searches for service information

The corresponding design class diagram :

- The interface layer has the jsp pages: ifMainCustomer, ifSearchService, ifDetailService
- The data manipulation layer has the classes Controller, ServiceController
- The data manipulation layer has the classes DAO, ServiceDAO
- Related entity classes

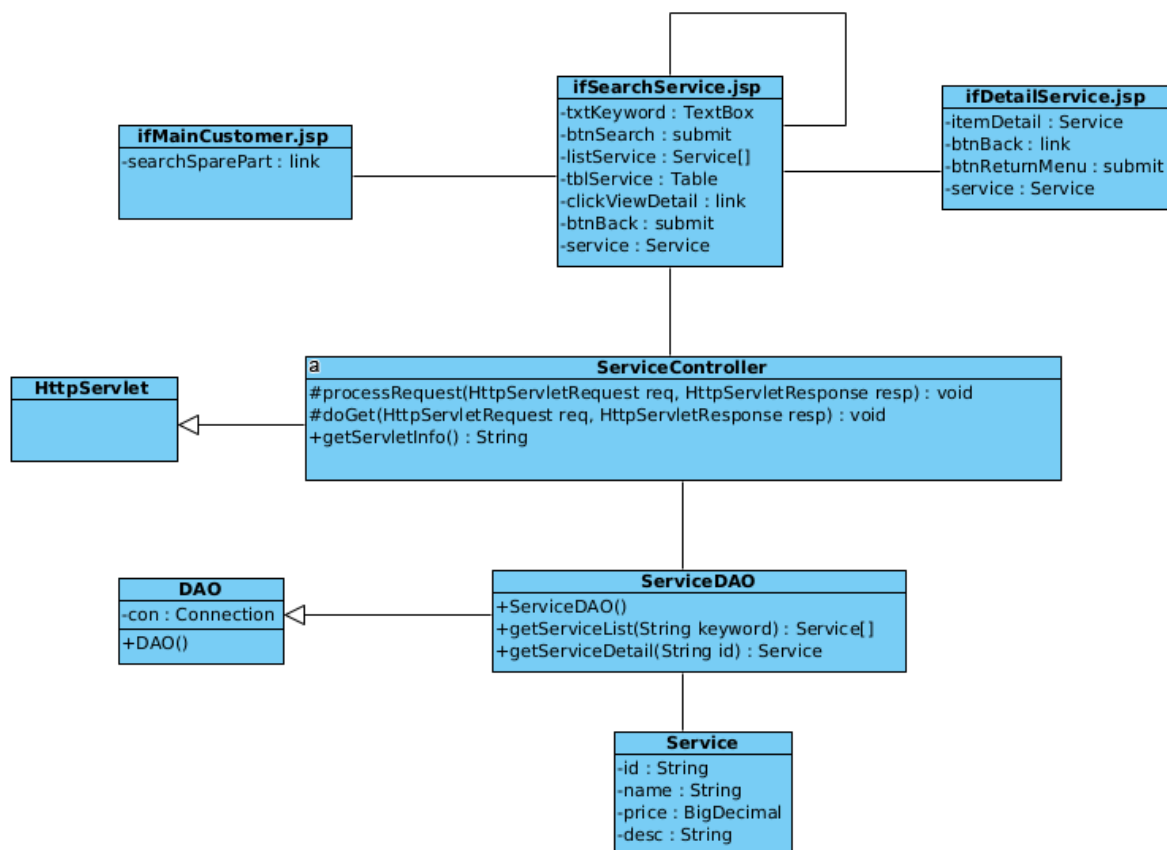


Figure 3.3. Class diagram for module search service

Usecase 2: Customer searches for spare parts information

The corresponding design class diagram :

- The interface layer has the jsp pages: ifMainCustomer, ifSearchSparePart, ifDetailSparePart
- The data manipulation layer has the classes DAO, SparePartDAO
- The data manipulation layer has the classes Controller, SparePartController
- Related entity classes

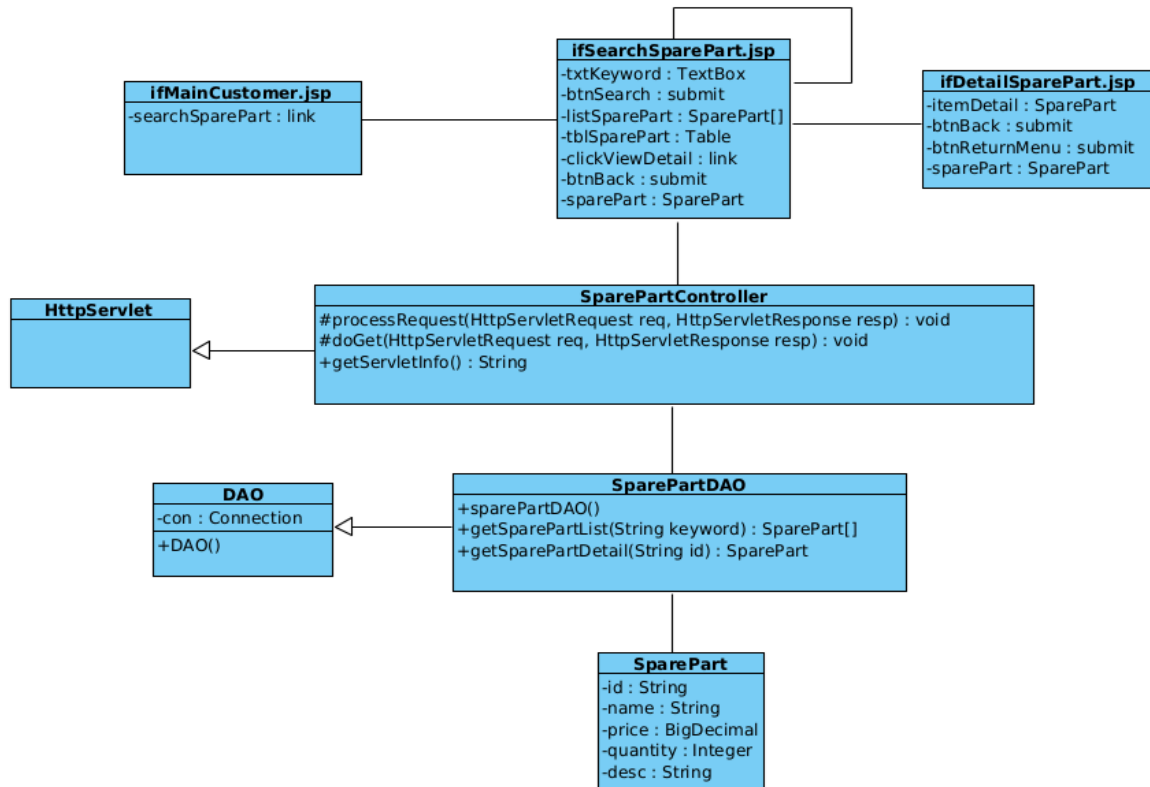


Figure 3.4. Class diagram for module search spare part

b. Module 2: Management staff views customer statistics

The corresponding design class diagram :

- The interface layer has the jsp pages: ifMainManagementStaff, ifSelectStatistics, ifViewCustomerStatistics, ifViewCustomerInvoices, ifInvoiceDetail
- The data manipulation layer has the classes Controller, CustomerStatController, PaymentInvoiceController
- The data manipulation layer has the classes DAO, CustomerStatDAO, PaymentInvoiceDAO
- Related entity classes

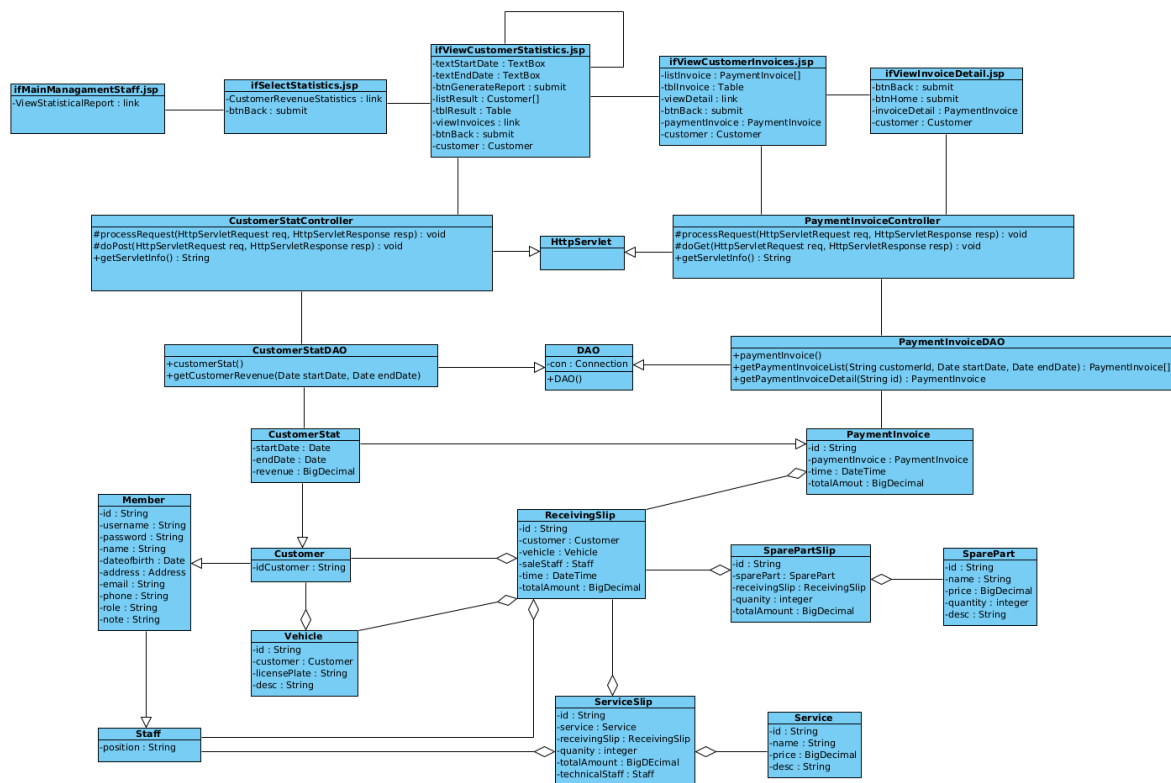


Figure 3.5. Class diagram for module view customer statistics

5. Activity diagram

a. Module 1: Customer searches for service/spare parts information

Usecase 1: Customer searches for service information

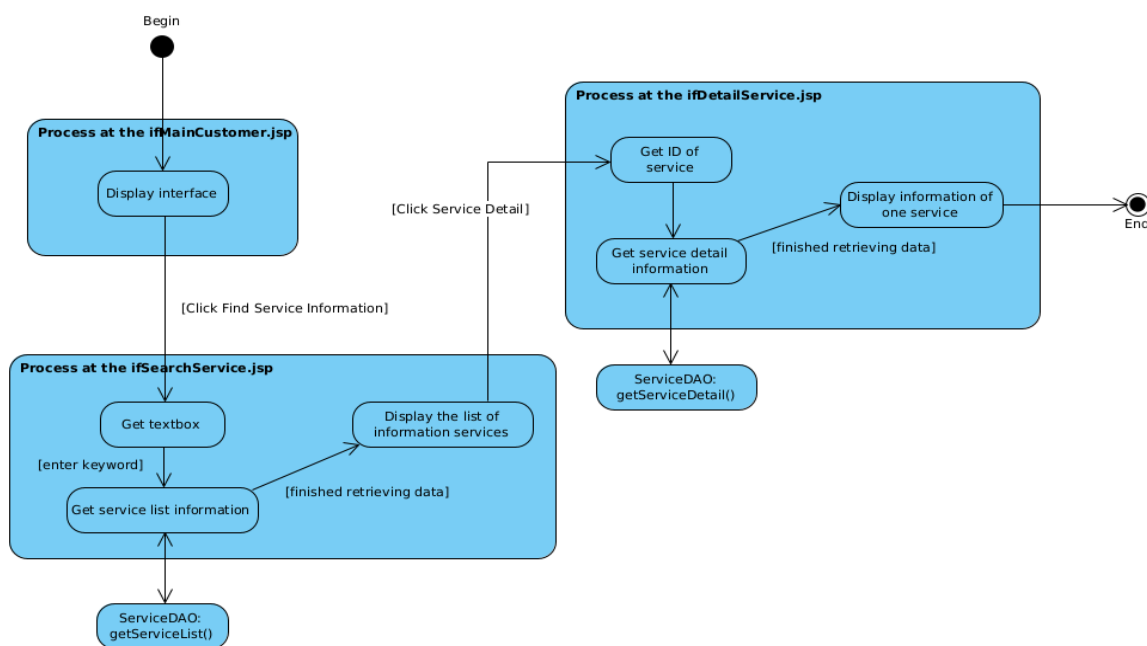


Figure 3.6. Activity diagram for module search service

Usecase 2: Customer searches for spare parts information

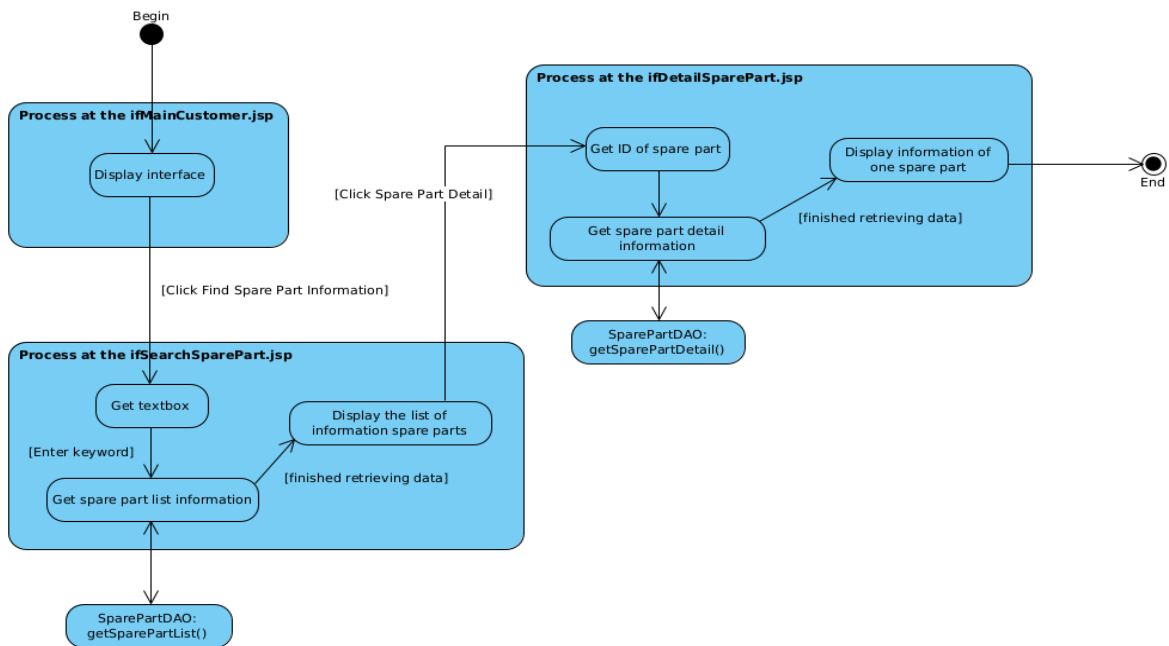


Figure 3.7. Activity diagram for module search spare part

b. Module 2: Management staff views customer statistics

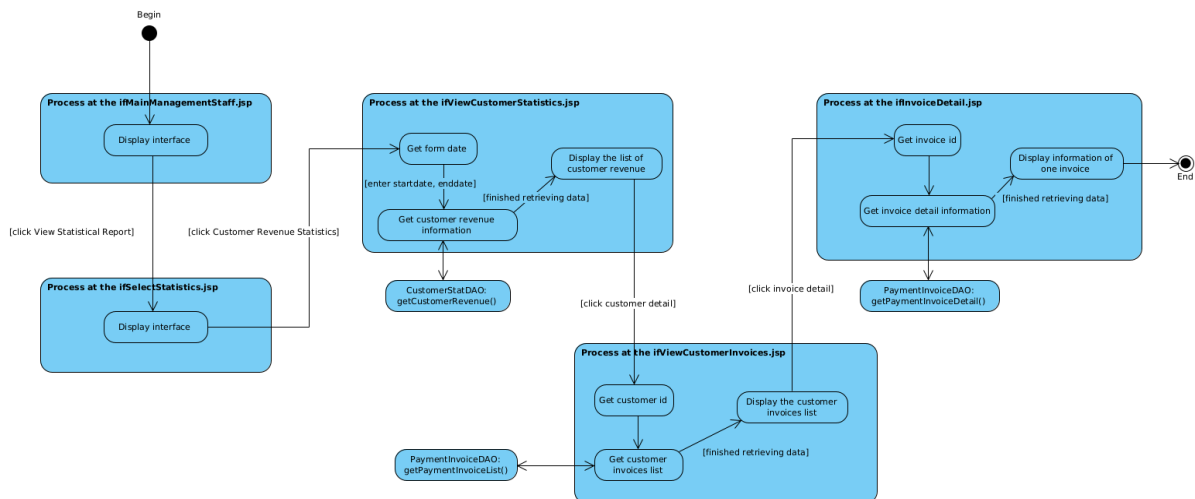


Figure 3.8. Activity diagram for module view customer statistics

6. Scenario 3.0

a. Module 1: Customer searches for service/spare parts information

Usecase 1: Customer searches for service information

1. On the main customer interface (ifMainCustomer.jsp), the customer clicks "Search Service".
2. The page ifMainCustomer.jsp calls ifSearchService.jsp.
3. The page ifSearchService.jsp displays the search form.
4. The customer enters a keyword and clicks Search.
5. The page ifSearchService.jsp calls the ServiceController.
6. The ServiceController executes the doGet() method.
7. The ServiceController calls the ServiceDAO.
8. The ServiceDAO calls the function getServiceList(keyword).
9. The function getServiceList(keyword) creates a new Service() object.
10. A Service object is initialized.
11. The Service object is returned to ServiceDAO.
12. The ServiceDAO returns the List<Service> to ServiceController.
13. The ServiceController returns the result to ifSearchService.jsp.
14. The page ifSearchService.jsp displays the service table.
15. The customer clicks one service in the table to view details.
16. The page ifSearchService.jsp calls the page ifServiceDetail.jsp
17. The page ifServiceDetail.jsp calls the ServiceController.
18. The ServiceController executes another doGet() request.
19. The ServiceController calls the ServiceDAO.
20. The ServiceDAO calls the function getServiceDetail(idService).
21. The function getServiceDetail(idService) creates a new Service() object.
22. A Service object is initialized.
23. The Service object is returned to ServiceDAO.
24. The ServiceDAO returns the Service detail to ServiceController.
25. The ServiceController returns the result to ifServiceDetail.jsp.
26. The page ifServiceDetail.jsp displays the service detail to the customer.

Usecase 2: Customer searches for spare parts information

1. On the main customer interface (ifMainCustomer.jsp), the customer clicks "Search Spare Part".
2. The page ifMainCustomer.jsp calls ifSearchSparePart.jsp.
3. The page ifSearchSparePart.jsp displays the search form.
4. The customer enters a keyword and clicks Search.
5. The page ifSearchSparePart.jsp calls the SparePartController.
6. The SparePartController executes the doGet() method.
7. The SparePartController calls the SparePartDAO.
8. The SparePartDAO calls the function getSparePartList(keyword).
9. The function getSparePartList(keyword) creates a new SparePart() object.
10. A SparePart object is initialized.
11. The SparePart object is returned to SparePartDAO.
12. The SparePartDAO returns the List<SparePart> to SparePartController.
13. The SparePartController returns the result to ifSearchSparePart.jsp.
14. The page ifSearchSparePart.jsp displays the spare part table.

15. The customer clicks one spare part in the table to view details.
16. The page ifSearchSparePart.jsp calls the page ifSparePartDetail.jsp
17. The page ifSparePartDetail.jsp call the SparePartController.
18. The SparePartController executes another doGet() request.
19. The SparePartController calls the SparePartDAO.
20. The SparePartDAO calls the function getSparePartDetail(idSparePart).
21. The function getSparePartDetail(idSparePart) creates a new SparePart() object.
22. A SparePart object is initialized.
23. The SparePart object is returned to SparePartDAO.
24. The SparePartDAO returns the SparePart detail to SparePartController.
25. The SparePartController returns the result to ifSparePartDetail.jsp.
26. The page ifSparePartDetail.jsp displays the spare part detail to the customer.

b. Module 2: Management staff views customer statistics

1. Management staff clicks “View Statistical Report” on ifMainManagementStaff.jsp.
2. ifMainManagementStaff.jsp calls ifSelectStatistics.jsp.
3. ifSelectStatistics.jsp displays available statistical categories.
4. Staff clicks “Customer Revenue Statistics”.
5. ifSelectStatistics.jsp calls ifViewCustomerStatistics.jsp.
6. ifViewCustomerStatistics.jsp displays interface with date range input and button.
7. Staff enters a date range and clicks the button.
8. ifViewCustomerStatistics.jsp calls CustomerStatController.
9. CustomerStatController executes doPost().
10. CustomerStatController calls CustomerStatDAO.
11. CustomerStatDAO executes getCustomerRevenue().
12. Inside getCustomerRevenue(), DAO queries the database to aggregate total revenue by customer within the date range.
13. Query results are encapsulated into CustomerStat objects.
14. CustomerStat objects return to CustomerDAO.
15. CustomerStatDAO returns the list of CustomerStat objects to CustomerStatController.
16. CustomerStatController returns the data to ifViewCustomerStatistics.jsp.
17. ifViewCustomerStatistics.jsp receives and processes the data, displays the list of customers with their total revenue.
18. Staff clicks a specific customer to view their invoices.
19. ifViewCustomerStatistics.jsp calls ifViewCustomerInvoices.jsp, passing idCustomer.
20. ifViewCustomerInvoices.jsp calls PaymentInvoiceController.
21. PaymentInvoiceController executes doGet().
22. PaymentInvoiceController calls PaymentInvoiceDAO.
23. PaymentInvoiceDAO executes getPaymentInvoiceList(idCustomer, dateFrom, dateTo).

24. Inside the method, DAO queries the database to retrieve all invoices for the selected customer in the date range.
25. Results are encapsulated into PaymentInvoice objects.
26. PaymentInvoice objects return to PaymentInvoiceDAO.
27. PaymentInvoiceDAO returns the list of PaymentInvoice objects to PaymentInvoiceController.
28. PaymentInvoiceController returns the list to ifViewCustomerInvoices.jsp.
29. ifViewCustomerInvoices.jsp displays the list of invoices for the selected customer.
30. Staff clicks a specific invoice to view details.
31. ifViewCustomerInvoices.jsp calls ifInvoiceDetail.jsp, passing idPaymentInvoice.
32. ifInvoiceDetail.jsp calls PaymentInvoiceController.
33. PaymentInvoiceController executes doGet().
34. PaymentInvoiceController calls PaymentInvoiceDAO.
35. PaymentInvoiceDAO executes getPaymentInvoiceDetail(idPaymentInvoice).
36. Inside the method, DAO queries the database to retrieve the detailed invoice record.
37. The result is encapsulated into a PaymentInvoice object.
38. PaymentInvoice object return to PaymentInvoiceDAO.
39. PaymentInvoiceDAO returns the PaymentInvoice object to PaymentInvoiceController.
40. PaymentInvoiceController returns the invoice detail to ifInvoiceDetail.jsp.
41. ifInvoiceDetail.jsp displays the detailed invoice information to the management staff.

7. Sequence diagram

a. Module 1: Customer searches for service/spare parts information

Usecase 1: Customer searches for service information

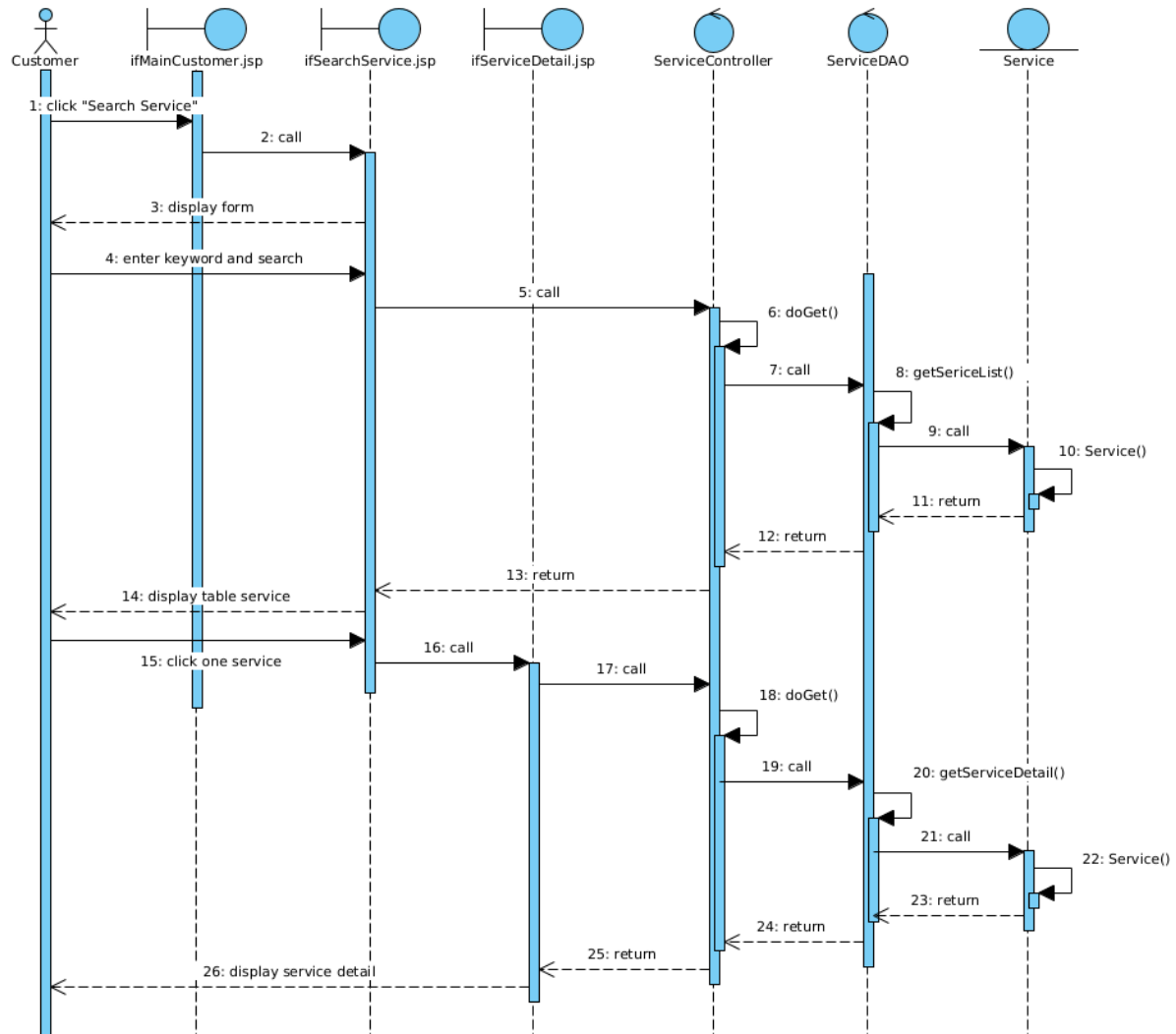


Figure 3.9. Sequence diagram for module search service

Usecase 2: Customer searches for spare parts information

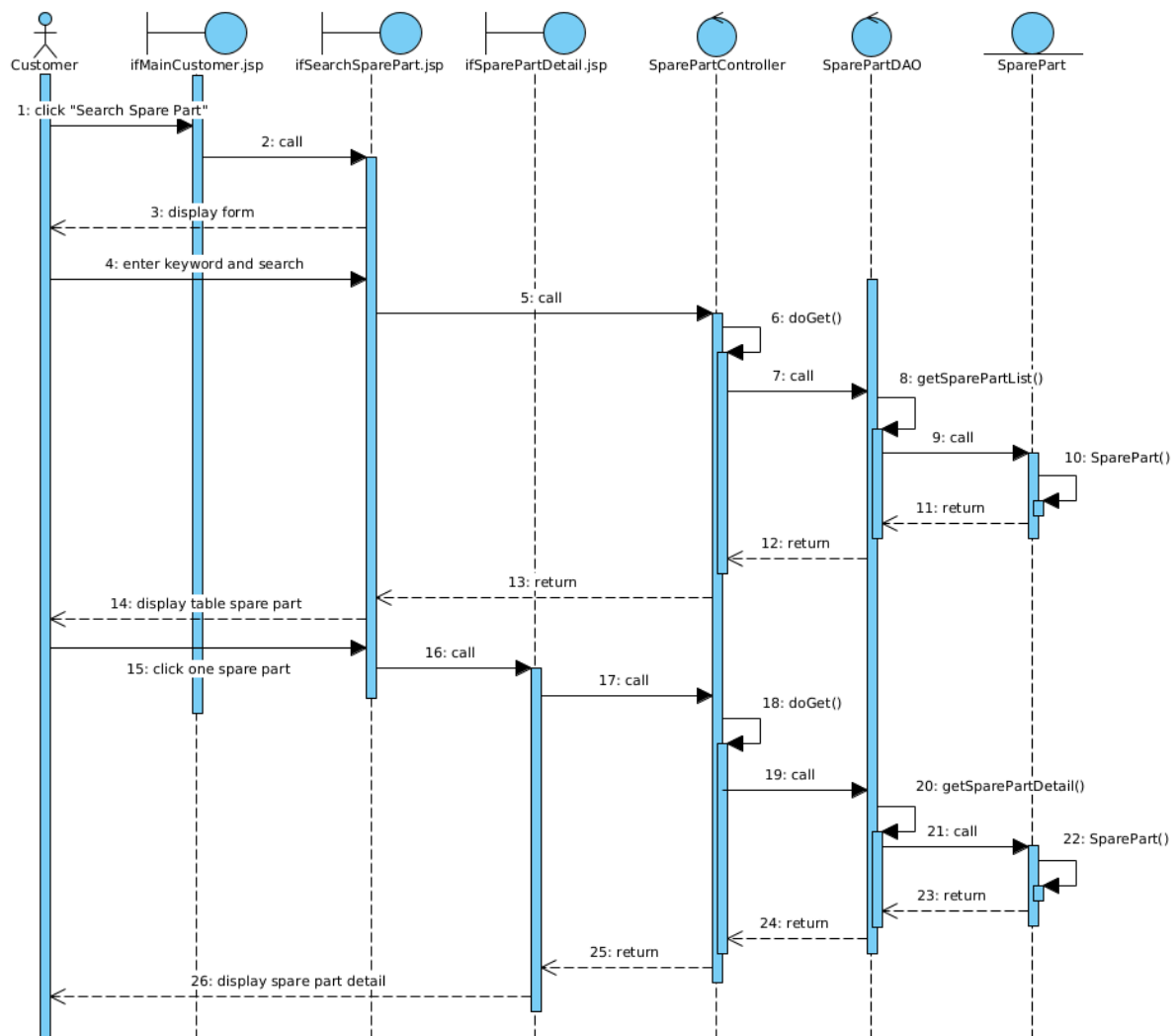


Figure 3.10. Sequence diagram for module search spare part

b. Module 2: Management staff views customer statistics

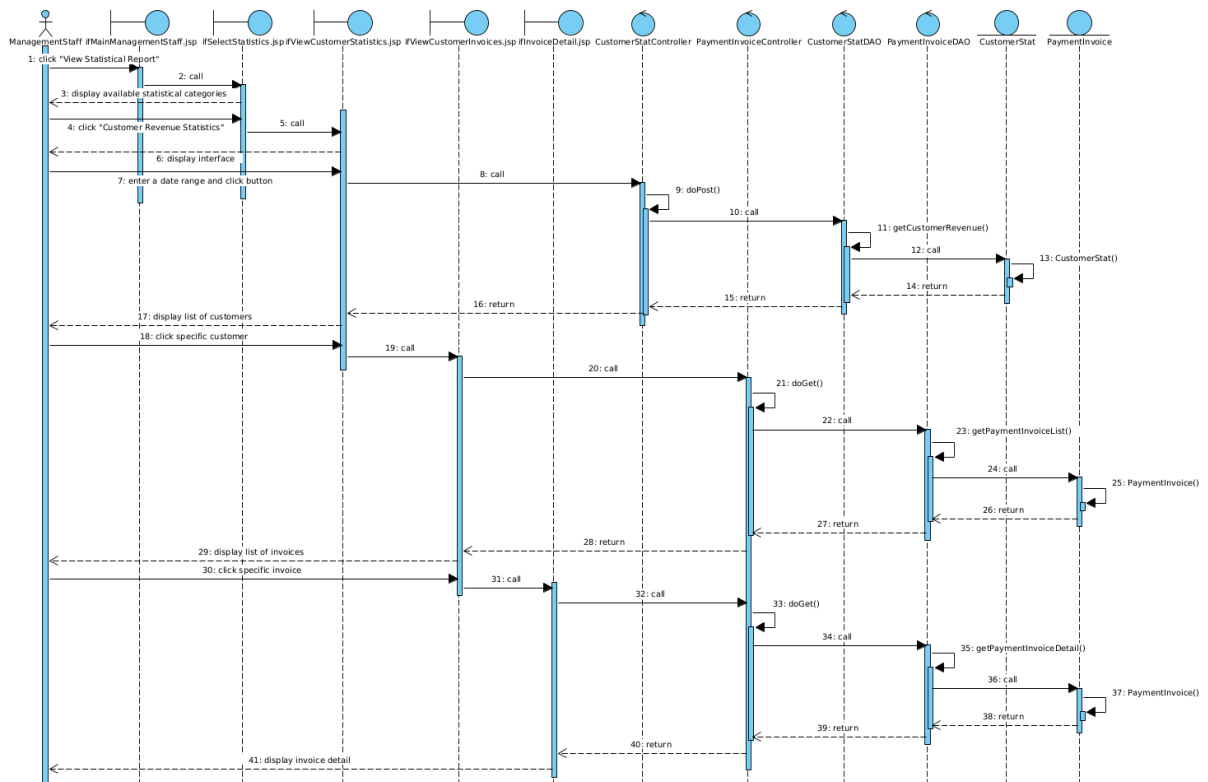


Figure 3.11. Sequence diagram for module view customer statistics

8. Package diagram

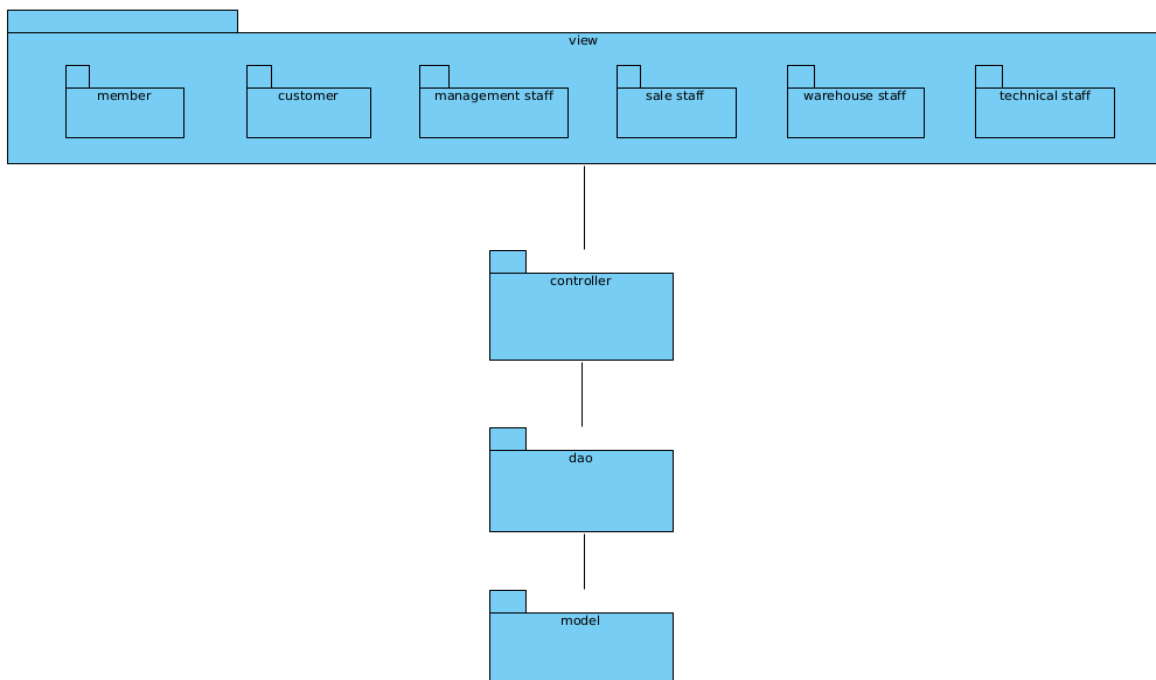


Figure 3.12. Package diagram

9. Deployment diagram

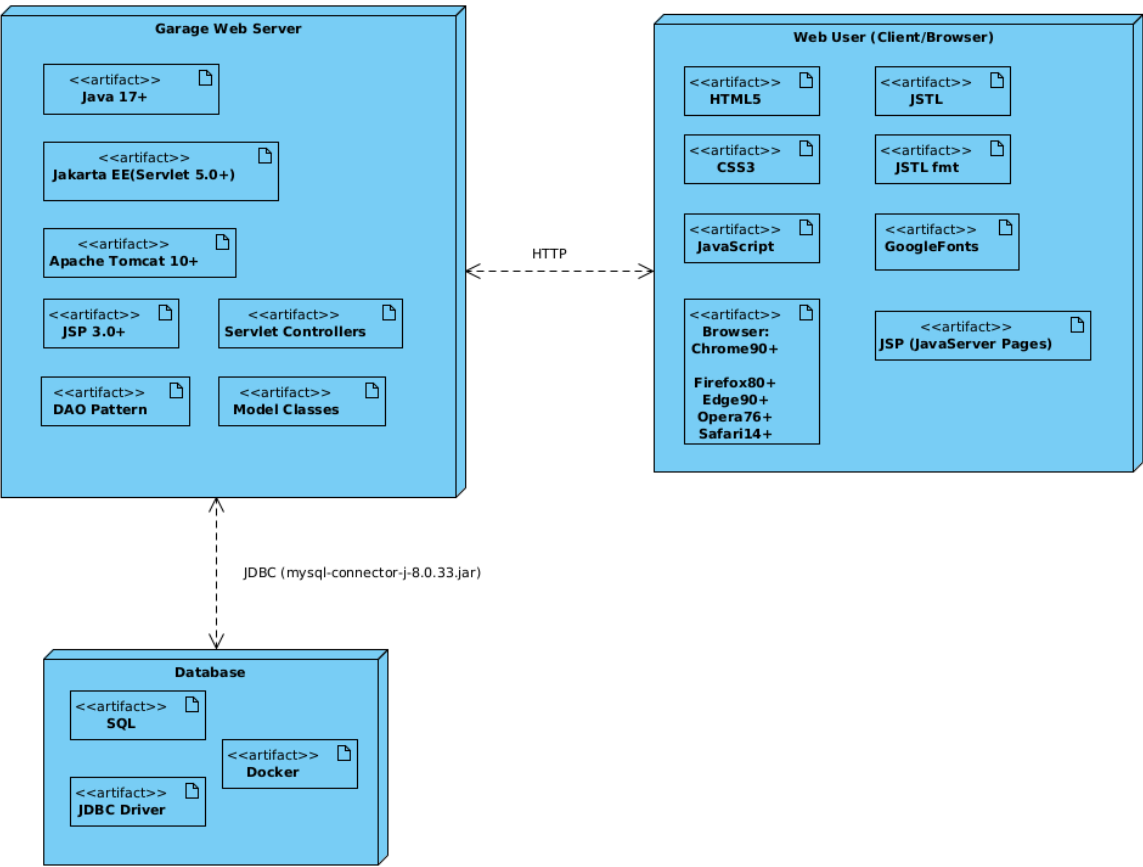


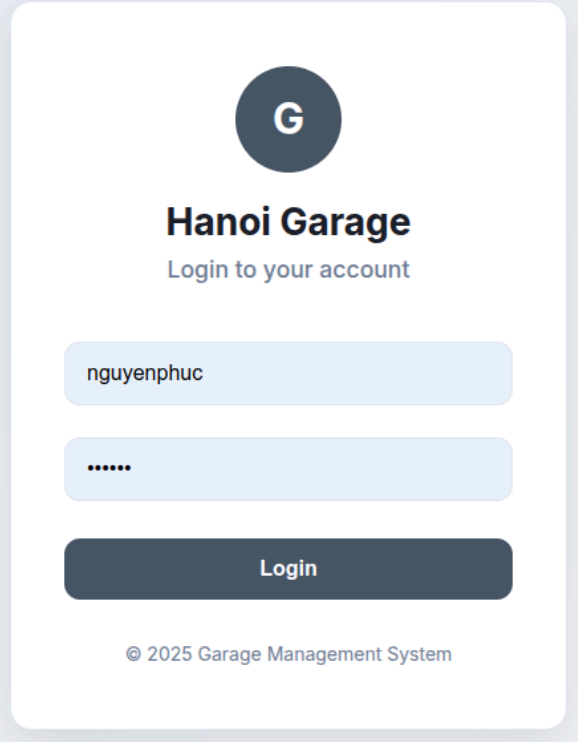
Figure 3.13. Deployment diagram

IV. Code

1. Interface

a. Module “Customer Searches for Service/Spare Parts Information”

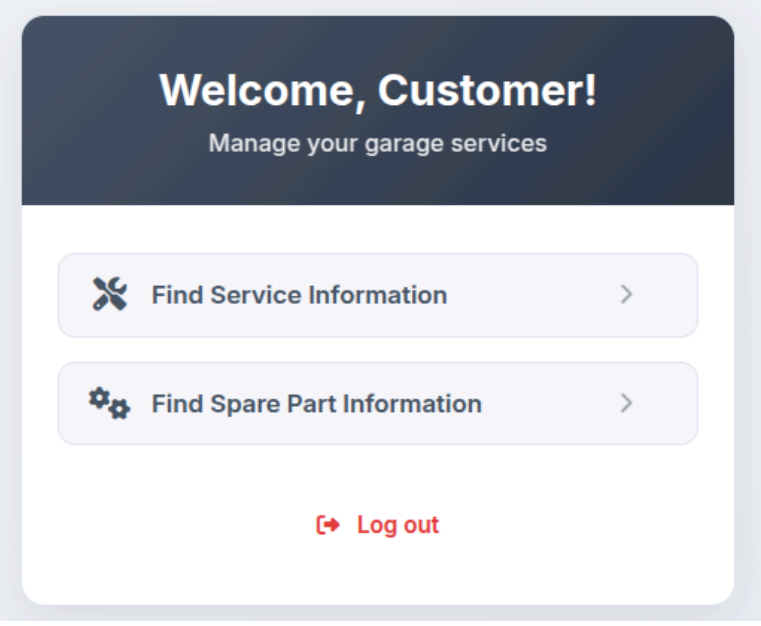
Displaying Login Interface



The login interface for Hanoi Garage. It features a dark blue circular logo with a white 'G' at the top. Below the logo is the text 'Hanoi Garage' in bold, followed by 'Login to your account' in a smaller font. There are two input fields: the first contains the username 'nguyenphuc' and the second contains masked characters '.....'. Below the input fields is a dark blue 'Login' button. At the bottom, there is a copyright notice: '© 2025 Garage Management System'.

After Customer login

=> Displaying **Main Customer Interface**



The main customer interface after login. It has a dark blue header with the text 'Welcome, Customer!' and 'Manage your garage services'. Below the header are two light blue buttons: 'Find Service Information' with a wrench icon and 'Find Spare Part Information' with a gear icon. At the bottom, there is a red 'Log out' button with a red arrow icon.

After Customer selects “Find Service Information”
=> Displaying **Search Service Interface**

Find Service Information

Service Catalog

Search

Search Spare Parts

ID	NAME	PRICE (VND)	ACTION
S1	Oil Change	200,000	View Details
S2	Tire Rotation	150,000	View Details
S3	Brake Inspection	250,000	View Details
S4	Battery Check	180,000	View Details
S5	Car Wash	100,000	View Details
S6	Air Filter Replacement	120,000	View Details
S7	Coolant Flush	350,000	View Details
S8	Transmission Fluid Change	450,000	View Details
S9	Spark Plug Replacement	280,000	View Details
S10	Wheel Alignment	300,000	View Details
S11	Windshield Wiper Replacement	90,000	View Details
S12	AC System Recharge	400,000	View Details
S13	Suspension Inspection	320,000	View Details

In **Search Service Interface**, Customer enter keyword and click Search
=> Displaying list of service matching the keyword

Find Service Information

Service Catalog

Search

Search Spare Parts

ID	NAME	PRICE (VND)	ACTION
S1	Oil Change	200,000	View Details
S8	Transmission Fluid Change	450,000	View Details
S20	Power Steering Fluid Change	240,000	View Details
S30	Differential Fluid Change	380,000	View Details
S53	Clutch Fluid Change	290,000	View Details

[Back to Home](#)

After Customer select specific service by click “View Details”
=> Displaying **Detail Service Interface**

Find Service Information

Service Detail

ID:

S1

Name:

Oil Change

Price:

200,000 VND

Description:

Engine oil replacement

[Back to List](#) | [Home](#)

After Customer selects “Find Spare Part Information”
=> Displaying **Search Spare Part Interface**

Find Spare Part Information

Spare Part Catalog

Search

Search Services

ID	NAME	PRICE (VND)	STOCK	ACTION
SP1	Engine Oil	100,000	100 pcs	View Details
SP2	Air Filter	80,000	50 pcs	View Details
SP3	Brake Pad Set	350,000	30 pcs	View Details
SP4	Battery 12V	1,200,000	20 pcs	View Details
SP5	Wiper Blade	150,000	60 pcs	View Details
SP6	Spark Plug Set	180,000	80 pcs	View Details
SP7	Oil Filter	120,000	90 pcs	View Details
SP8	Fuel Filter	95,000	70 pcs	View Details
SP9	Timing Belt	850,000	25 pcs	View Details
SP10	Clutch Kit	2,500,000	15 pcs	View Details
SP11	Radiator	3,200,000	12 pcs	View Details
SP12	Water Pump	680,000	35 pcs	View Details
SP13	Alternator	3,800,000	18 pcs	View Details
SP14	Starter Motor	2,200,000	22 pcs	View Details
SP15	Shock Absorber Front	950,000	40 pcs	View Details
SP16	Shock Absorber Rear	850,000	45 pcs	View Details
SP17	Brake Disc Front	750,000	30 pcs	View Details
SP18	Brake Disc Rear	650,000	35 pcs	View Details

In **Search Spare Part Interface**, Customer enter keyword and click Search
=> Displaying list of spare parts matching the keyword

Find Spare Part Information

Spare Part Catalog

Search

Search Services

ID	NAME	PRICE (VND)	STOCK	ACTION
SP1	Engine Oil	100,000	100 pcs	View Details
SP7	Oil Filter	120,000	90 pcs	View Details
SP33	Ignition Coil	580,000	42 pcs	View Details
SP46	Transmission Oil	180,000	80 pcs	View Details
SP47	Differential Oil	220,000	70 pcs	View Details

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After Customer select specific spare part by click “View Details”
=> Displaying **Detail Spare Part Interface**

Find Spare Part Information

Spare Part Detail

ID: SP1

Name: Engine Oil

Price: 100,000 VND

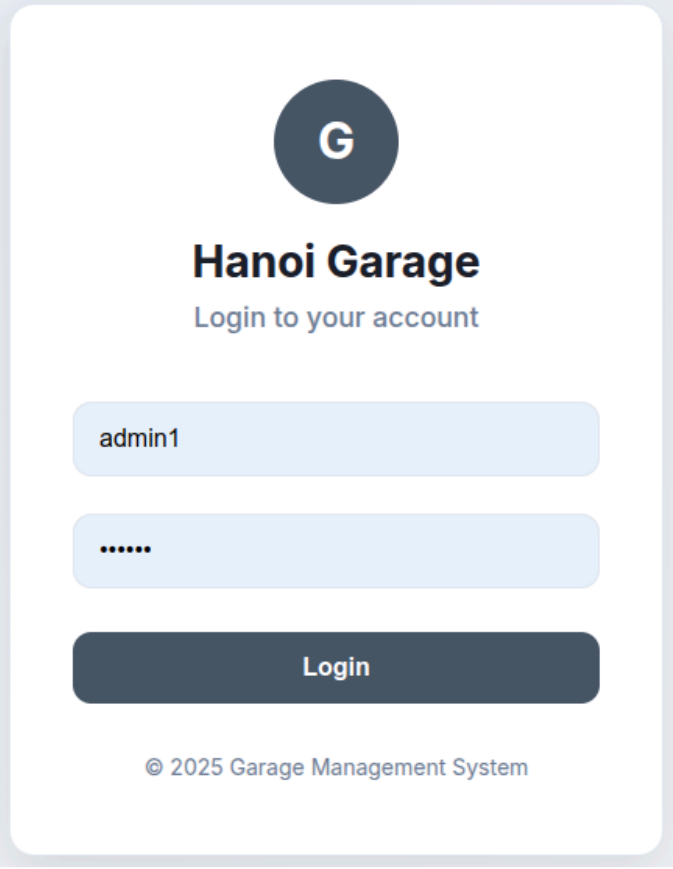
Stock: 100 pcs

Description: Synthetic 5W-30 oil

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b. Module 2: Management staff views customer statistics

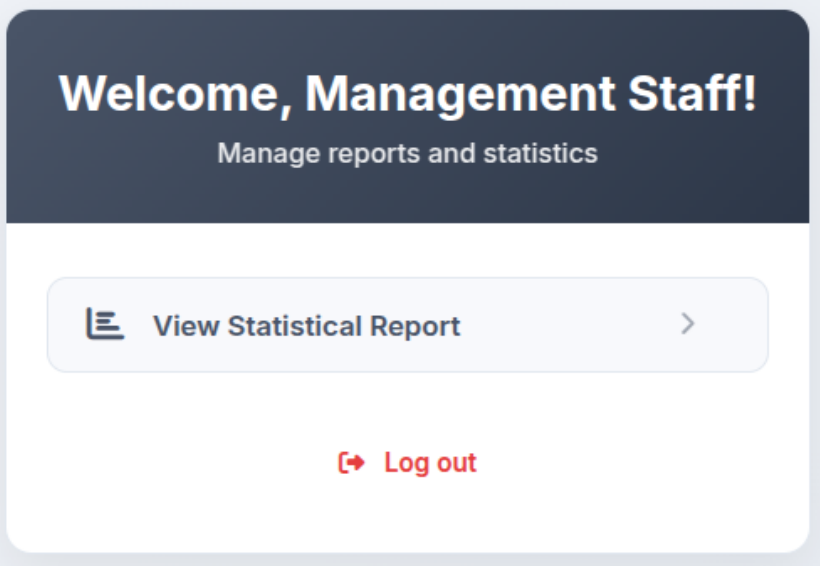
Displaying Login Interface



The login interface for Hanoi Garage. It features a dark blue circular logo with a white 'G' at the top. Below the logo, the text 'Hanoi Garage' is displayed in a bold, dark blue font, followed by 'Login to your account' in a smaller, lighter blue font. There are two light blue input fields: the first contains the text 'admin1', and the second contains six dots, indicating a password field. Below these fields is a dark blue 'Login' button. At the bottom, there is a copyright notice: '© 2025 Garage Management System'.

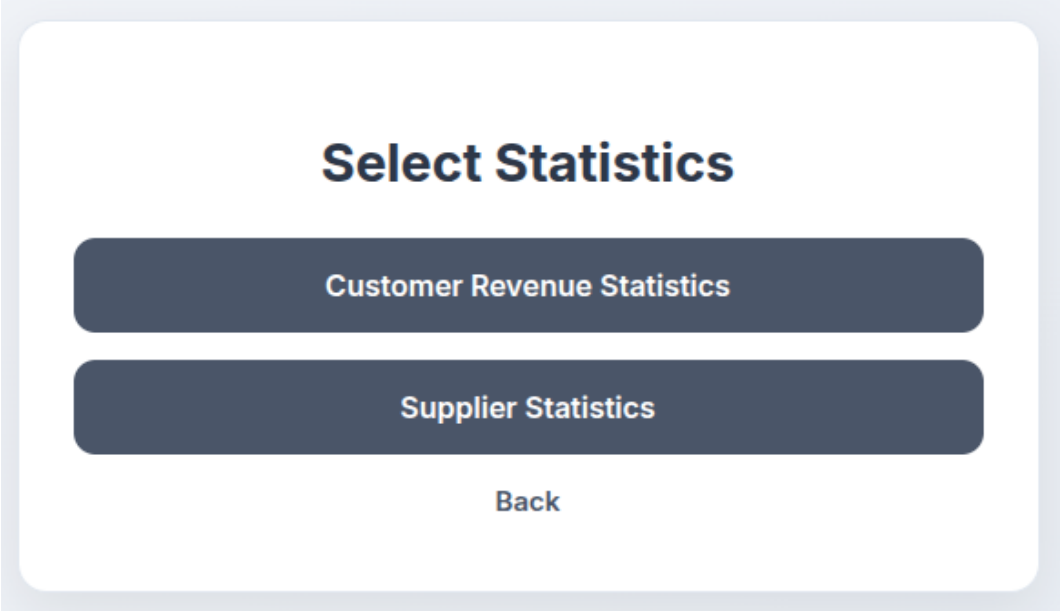
After Management Staff login

=> Displaying **Main Management Staff Interface**



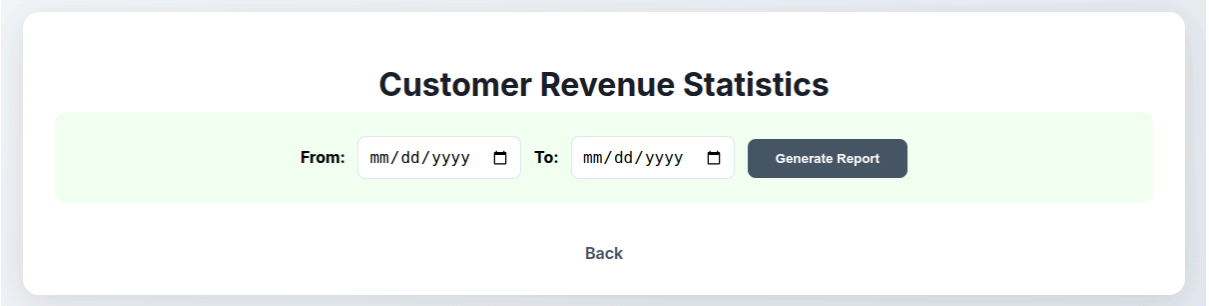
The main management staff interface. It has a dark blue header bar with the text 'Welcome, Management Staff!' in white, bold font, and 'Manage reports and statistics' in a smaller white font below it. Below the header, there is a light blue button with a document icon, the text 'View Statistical Report', and a right-pointing chevron. At the bottom, there is a red 'Log out' button with a red icon of a door with an arrow pointing out.

After Management Staff selects “View Statistical Report”
=> Displaying **Select Statistics Interface**



The interface is titled "Select Statistics" in a large, bold, dark blue font. Below the title, there are two large, dark blue buttons with white text. The first button is labeled "Customer Revenue Statistics" and the second button is labeled "Supplier Statistics". At the bottom center, there is a smaller, dark blue button labeled "Back".

After Management Staff selects “Customer Revenue Statistics”
=> Displaying **View Customer Statistics Interface**



The interface is titled "Customer Revenue Statistics" in a bold, dark blue font. Below the title, there is a light green horizontal bar containing two date input fields. The first field is labeled "From:" and the second field is labeled "To:". Both fields have a placeholder "mm/dd/yyyy" and a calendar icon. To the right of the date fields is a dark blue button labeled "Generate Report". At the bottom center, there is a smaller, dark blue button labeled "Back".

In the **View Customer Statistics Interface**, Management Staff enters date range and click Generate Report
=> Displaying the the list of customers with revenue

Customer Revenue Statistics

Period: 01/10/2025 to 30/11/2025

From:10/01/2025

To:11/30/2025

Generate Report

Customer ID	Name	Invoices	Revenue (VND)	Action
M16	Vo Minh Tuan	2	7,580,000	View Invoices
M12	Pham Van Dong	3	4,820,000	View Invoices
M2	Nguyen Huu Phuc	5	4,220,000	View Invoices
M9	Tran Van Nam	2	3,220,000	View Invoices
M8	Nguyen Van Hung	3	2,510,000	View Invoices
M15	Pham Gia Han	1	2,380,000	View Invoices
M24	Phan Thi Kim Oanh	2	2,380,000	View Invoices
M14	Le My Hoa	3	2,350,000	View Invoices
M31	Luong The Dung	1	2,350,000	View Invoices
M30	Kieu Van Dai	1	2,140,000	View Invoices
M23	Trinh Hoai Nam	2	1,920,000	View Invoices
M22	Ly Hoang Phuc	1	1,800,000	View Invoices
M17	Dang Thi Lan	2	1,610,000	View Invoices
M29	Hoang Yen Chi	2	1,500,000	View Invoices
M27	Nguyen Thao My	2	1,325,000	View Invoices
M21	Duong Van Long	2	1,295,000	View Invoices
M25	Mai Anh Tuan	1	1,280,000	View Invoices
M32	To Ngoc Diep	2	1,050,000	View Invoices
M20	Ngo Thanh Thuy	2	600,000	View Invoices
M10	Le Van Hoang	2	490,000	View Invoices
M18	Bui The Vinh	1	450,000	View Invoices
M26	Do Gia Bao	2	400,000	View Invoices
M28	Vu Duc Minh	1	380,000	View Invoices
M19	Ho Ngoc Ha	1	320,000	View Invoices
M13	Tran Van Anh	1	250,000	View Invoices

Total Revenue: 48,620,000 VND

Back

After Management Staff clicks specific customer in action “View Invoices”
=> Displaying **View Customer Invoices Interface**

Customer Revenue Statistics

Invoice History

Customer: Nguyen Huu Phuc

Revenue: 4,220,000 VND

Period: 01/10/2025 to 30/11/2025

ID	Date	Total Amount (VND)	Type	Action
PI1	01/10/2025	1,180,000	Card	View
PI7	04/10/2025	1,530,000	Bank Transfer	View
PI20	09/10/2025	150,000	Cash	View
PI31	01/11/2025	950,000	Cash	View
PI41	17/11/2025	410,000	Cash	View

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After Management Staff clicks specific invoice in action “View”
=> Displaying **Invoice Detail Interface**

Customer Revenue Statistics

Invoice History

Invoice ID: #PI1

Customer: Nguyen Huu Phuc

Staff: Le Van Huong

Vehicle: 51H-12345

Payment Method: Card

Date: 01/10/2025 09:00

Services

Name	Quantity	Total
Oil Change	1	200,000 VND
Air Filter Replacement	1	120,000 VND
Cabin Air Filter Replacement	1	110,000 VND

Spare Parts

Name	Quantity	Total
Engine Oil	4	400,000 VND
Oil Filter	1	120,000 VND
Cabin Air Filter	1	150,000 VND
Air Filter	1	80,000 VND

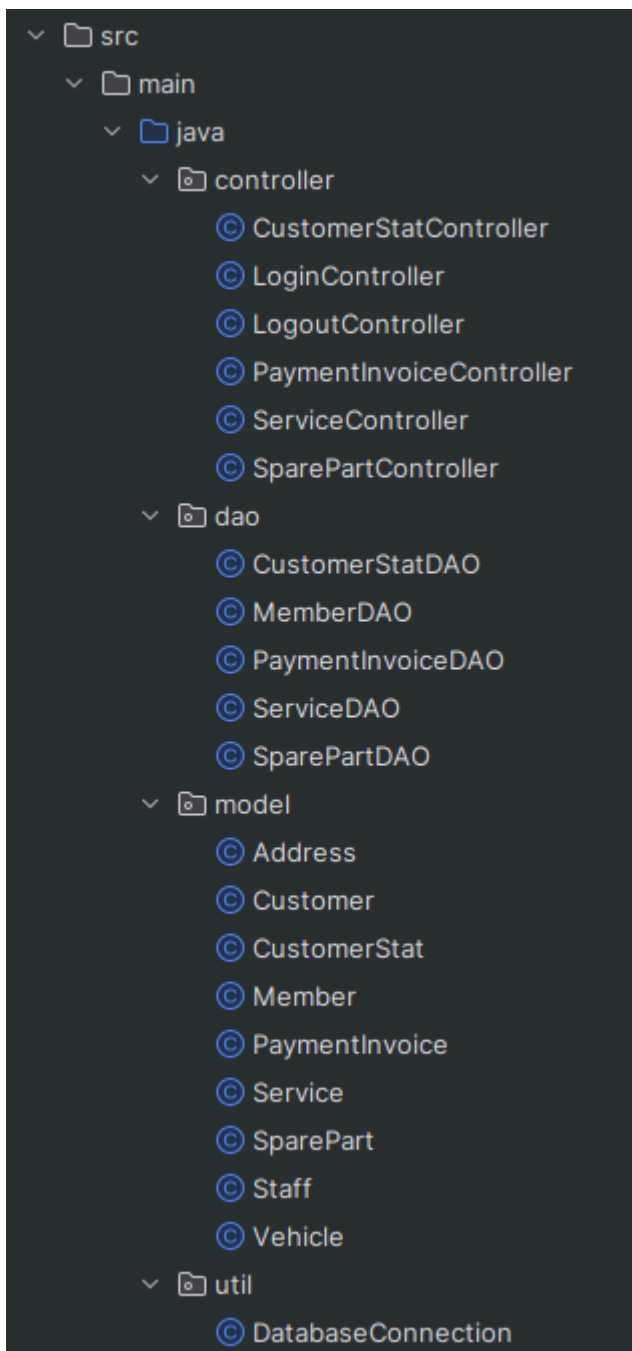
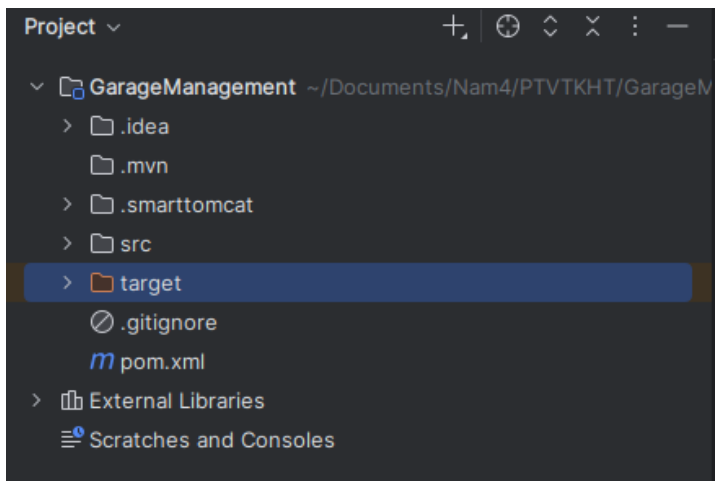
TOTAL: 1,180,000 VND

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2. File structure

Source code Github: [Link code Garage Management System](#)

Source code Zip: [File zip drive](#)



- ▼  webapp
 - ▼  Login
 - JSP ifMainCustomer.jsp
 - JSP ifMainManagementStaff.jsp
 - JSP login.jsp
 - ▼  SearchService
 - JSP ifDetailService.jsp
 - JSP ifSearchService.jsp
 - ▼  SearchSparePart
 - JSP ifDetailSparePart.jsp
 - JSP ifSearchSparePart.jsp
 - ▼  StaffViewStatistics
 - JSP ifInvoiceDetail.jsp
 - JSP ifSelectStatistics.jsp
 - JSP ifViewCustomerInvoices.jsp
 - JSP ifViewCustomerStatistics.jsp
 - ▼  WEB-INF
 - </> web.xml
 - JSP index.jsp